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DESIGN PRINCIPLES FOR LIFTING SURFACES COVERS WITH INTEGRATED STIFFENERS MANUFACTURED WITH LIQUID RESIN INFUSION (LRI) AND AUTOMATIC FIBRE PLACEMENT (AFP)

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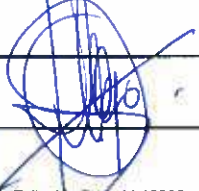
Summary / Resumen

Nowadays, prepreg based technology is the most common manufacturing process to produce high performance composites. However, it presents some disadvantages, such as high investment in autoclaves and cold storage, low level of integration and long processing cycles. Direct processes can overcome these disadvantages, combining and curing resin and reinforcement at the same time. In addition, LRI also offers the possibility to produce larger parts with lower associated costs than RTM.

In this Master Thesis of Master in Composites, design principles for stiffened lifting surfaces covers manufactured with AFP and LRI will be described. They are going to be obtained from existing test specimens and manufacturing demonstrators.

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1. INTRODUCTION

1.1 Scope of the project

A new external wing based on Airbus Defence & Space C295 model is being developed in the European Union R&T program Clean Sky 2. In Step 2 the external composite wing box is being manufactured promoting composite green manufacturing and out of autoclave processes. The lower half is being manufactured with LRI, including skin, stringer and front and rear spars by one shot.

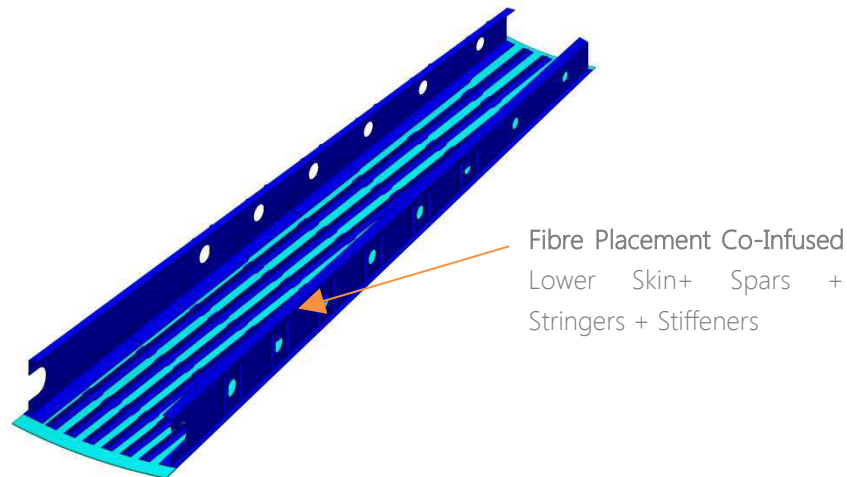


Figure 1 External Composite Wing box Clean Sky 2 (lower half)

Based on the experience and lessons learnt from this project, the objectives of this master thesis of Master in Composites are:

- 1) To understand LRI technology and the impacts it has in structural design
- 2) To obtain general tolerances from dimensional studies of test specimens
- 3) To provide design principles applicable to composite parts manufactured with Automatic Fibre Placement (AFP) and Liquid Resin Infusion (LRI), specifically for lifting surfaces covers with integrated stiffeners

1.2 Content

In this document, the design principles for composite parts manufactured with AFP and LRI will be explained, being the more generic as possible, but also providing specific design guidelines. For that reason the second part corresponding to design principles section is divided into two parts: general design principles and specific design principles.

Furthermore a dimensional study obtained from reports of existing test specimens has been made, with which the design principles exposed can be verified, the general manufacturing tolerances expected and the possibilities of this emerging technology.

1.3 Acronyms

LRI: liquid resin infusion

AFP: automatic fibre placement

RTM: resin transfer moulding

UD: unidirectional

CS2: Clean Sky 2

GFRP: Glass Fibre Reinforced Plastic

NDT: non-destructive test

MCL: minimum cut length

TBD: to be determined

MKDF: manufacturing key design feature

1.4 References

Reference	Document	Issue	Title
Ref. 1	AIPS03-02-022	10	Manufacture of Monolithic and Sandwich Structural Components by impregnation of dry reinforcement materials with Thermoset Resin
Ref. 2	TAE-CS-MM-150002	A	Clean Sky 2 outer external wing baseline solution Preliminary sizing of skin covers
Ref. 3	TAE-CS-MM-160001	A	Clean Sky 2 outer external wings spars preliminary sizing. Baseline solution
Ref. 4	ABS5797	3	General Manufacturing tolerances for non-metallic composite components of “new generation composite aircraft.
Ref. 5	MIL-STD1472G		Department of defense Design Criteria Standard. Human engineering

Ref. 6	ABD0078	D	Corrosion prevention
Ref. 7	TAE-CS-DT-160002	B	Test CS A-1.3 Outer Wing Box- lower cover. Lightning Strike test plan
Ref. 8	CAN11055	9	Dimensional Tolerances for Composite Material Parts
Ref. 9	V51RSDP07001.2	2.0	Reference Structure Design Principles for A350XWB volume 2 General Design Principles
Ref. 10	V51RSDP07001.5	3.0	Reference Structure Design Principles for A350XWB Volume 5 Wing Design Principles
Ref. 11	Hexcel Manual		Direct Processes Technology Manual
Ref. 12	Hexcel Data Sheet		Hexflow RTM6
Ref. 13	Hexcel Data Sheet		Hitape Reinforcements
Ref. 14	Hexcel article		HiTape: A new efficient composite technology for Primary Aircraft Structures
Ref. 15	ESD-UG0300178	1	A400M Composite Design Principles
Ref. 16	TAE-CS-NT-160002	A	General Technical Requirements for Clean Sky 2 FTB2 Structural Components
Ref. 17	D-B-2.2.1-23	WP 5.2	Critical Composite Wing Build Process

2. LIQUID RESIN INFUSION MANUFACTURING

2.1 Background: direct processes

Traditional composites manufacturing processes, which are widely used, involve prepreg and the use of autoclaves and require high investment in cold storage and autoclaves.

LRI is a direct process in which resin is infused into a reinforcement preform, placed in a one-sided mould with bagging film. There is an existing document (see Ref. 1) in which some particularities of the process are explained.

In direct process the resin and the reinforcement are combined and cured at the same time, so the cold storage is not necessary and the shipping costs are considerably reduced. Autoclaves are not part of the process (they are also called out of autoclave processes). Therefore, a lot of investment and energy are saved.

Furthermore, having resin and reinforcement separated also offers a great advantage: recycling is much easier. Both materials are separated until the final curing process. Consequently, until this curing process, the scrap of reinforcement, which is the most expensive part, can be recycled without the necessity and difficulty of separating it from the resin. Also the styrene emissions due to resin curing are minimized, because the process is effectuated in a closed environment.

Apart from these economic and environmental advantages, direct process offers others, such as:

- Higher possibility of integration: the process indeed allows the manufacturing of much more complex parts than with prepreg technology, so more parts can be integrated in one curing cycle. This means no riveting or adhesive is needed to assemble the parts.
- The scrap is substantially reduced.
- Processing cycles are considerably shorter than autoclave. Also less assembly steps are needed.
- Adequate for high production rate.
- The surface finish is better in comparison with prepreg technology.

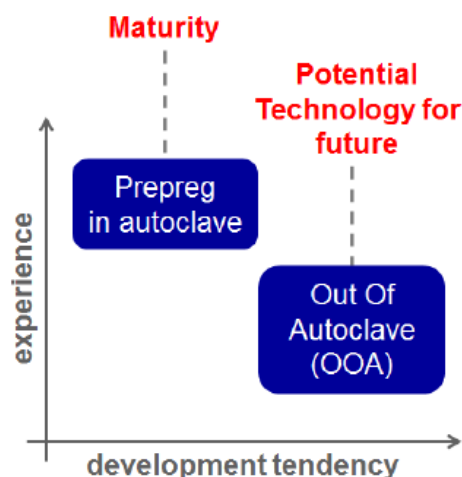


Figure 2 Tendency of out of autoclave processes (from Ref. 11)

Processes in autoclaves are highly industrialised, because they have been widely used and there is considerable experience in this technology. However, the advantages that direct processes offer are motivating the development tendency of these technologies.

In summary, direct processes have a future potential due to their multiple advantages over prepreg and autoclave processes.

2.2 Liquid Resin Infusion

LRI processes also have some additional advantages over RTM:

- The necessary tooling is simpler, as only one mould is needed and it does not have to be matched with another, so the investment on moulds is lower.
- It is more adequate for large surface components

In the next figure a scheme for LRI equipment is shown:

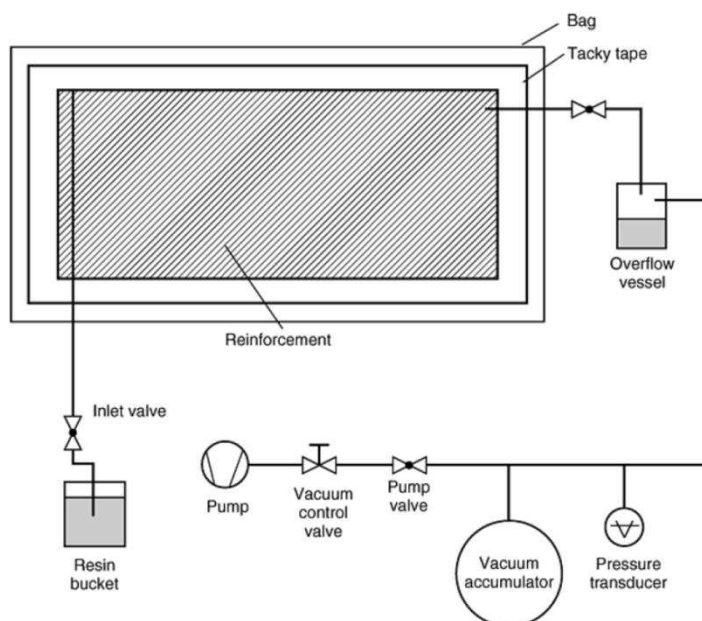


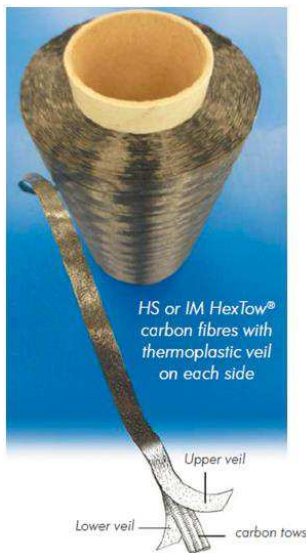
Figure 3 LRI equipment scheme

It should include a Programmable Logic Controller to control the quantity of resin it injects (g/min).

2.3 Materials

The primary materials that have been used for these parts for test specimens are:

- Reinforcement: Hexcel Hi-Tape UD210 IMA-12K/12.7mm/V800E
- Resin: RTM6 (AIMS 05-04-100, IPS 05-04-100-01)



Material allowables have been obtained from coupon test and can be consulted in Ref. 2.

Both are materials from Hexcel. The information of these materials here displayed has been collected from Hexcel specification sheets.

Figure 4 Hexcel Hi-tape(from Ref. 13)

-Reinforcement:

Hexcel Hitape UD210: this dry unidirectional reinforcements, called hi-tape, has been designed to out of autoclave technologies, like LRI or RTM. Its lay-up can be fully automated like prepreg. Its fibre volume content and mechanical properties are similar to UD prepreg. In addition it has higher Compression after Impact (CAI) performance than other materials used in LRI. For coupon tests the tape width used was 6.35 mm (1/4 inch), but for test specimens the width used has been 12.70 mm (1/2 inch). For the design of the outer external wing box lower skin 12.70 tape width is also being used.

The thickness considered per cured ply is 0.196 mm.

As it can be seen in the previous figure, the reinforcement has a thermoplastic veil on each side.

Width versus areal weight								
Width in mm (inch)	HexTow® AS7 Areal weight in gsm			HexTow® IMA/IM7 Areal weight in gsm				
6.35 (1/4)	126	-	252	140	-	210	-	280
12.70 (1/2)	126	190	252	140	175	210	245	280
Up to 500 (20)	Standard areal weights							

Table 1 Hexcel Hi-tape physical properties (from Ref. 13)

-Resin:

Hexflow RTM 6: is the adequate thermoset resin to obtain the optimum mechanical performances for the previous material. It is a widely used resin with proven operation. One of its advantages is high glass transition temperatures and flexibility in injection and cure cycle parameters. It is specifically convenient for direct processes because it is easy to process, low injection pressure is needed. It can be stored at – 18°C during 6 months.

Neat Resin Properties

	Tensile	Flexure
Strength (MPa)	75	132
Modulus (MPa)	2890	3300
Strain (%)	3.4	
Standard specifications	ASTM D638	ASTM D790

Uncured resin density: 1.11 g/cm³ at 25°C
 Cured resin density: 1.14 g/cm³ at 25°C
 Fracture toughness (G1C /ASTM D 5045): 89 J/m²
 Coefficient of Thermal Expansion: 52.7 e-6 /K

Table 2 Hexcel RTM6 resin properties (from Ref. 12)

Apart from the reinforcement and the resin, two more materials may be included in the laminate when required:

- **Glass fibre:** hexcel ABS5382, B type. Used for metallic interface (as it will be explained in following sections). Its thickness is 0,1 mm.
- **Bronze mesh:** Z22946-ABS5327A03. Standard bronze mesh with a new coat (specifically developed to be integrated with infusion material). Used for zone 2A (as it will be explained in following sections). Its mass is 80 g/m².

2.4 Process steps

In the next scheme it is shown a general manufacturing process for the reinforcement used:

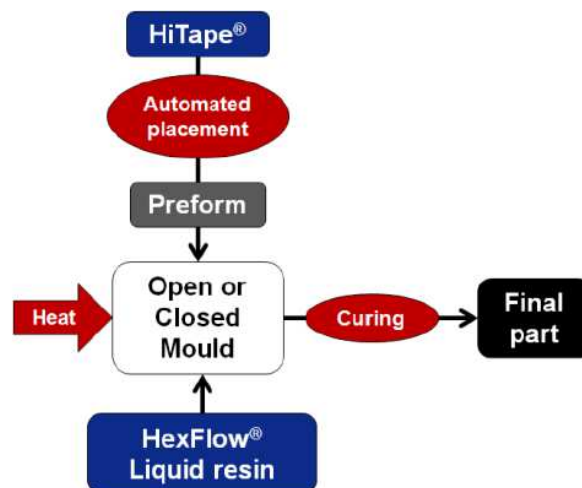


Figure 5 Manufacturing process with Hexcel Hi-tape (from Ref. 14)

For this specific example of CS2 project, there are some particularities in the manufacturing process, due to the level of development of the technologies and the availability of machines for this specific program. The process steps are being described with these particularities:

- **Lay-up:** the reinforcement is placed using a numerical controlled head and compacted by a roller onto a rigid tool surface with the AFP to create the preform.

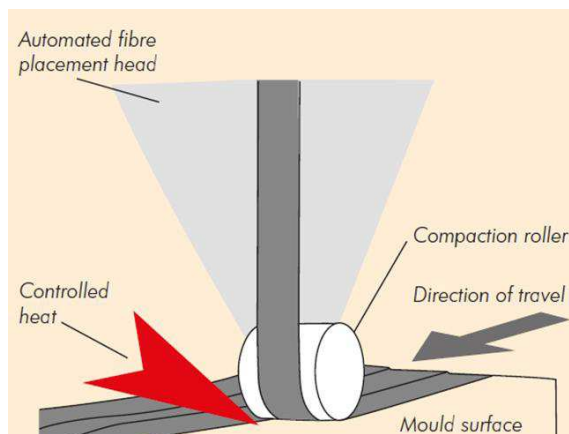


Figure 6 AFP typical scheme (from Ref. 14)

In this specific case it has been laminated in a vertical surface, although the most common is using a horizontal surface. For the first ply saerfix has been used (its use will be explained in following sections).

The manipulation of this material should be done in clean area with controlled temperature and humidity.



Figure 7 AFP FIDAMC

- **Hot-forming:** the preform is placed in a press where pressure and temperature are applied to form the corresponding shape. Temperature can be applied by radiation, conduction or convection.

The temperature of the hot-forming cycle rises to 170°C in 60 minutes. After this time the stabilization time and vacuum begins. From minute 65 vacuum is constant until minute 155, where cooling begins until the temperature decreases to 60°C, which is considered room temperature, and the hot-forming cycle finishes. It is important to reach these temperatures due to the thermoplastic veil that the reinforcement has on each side.

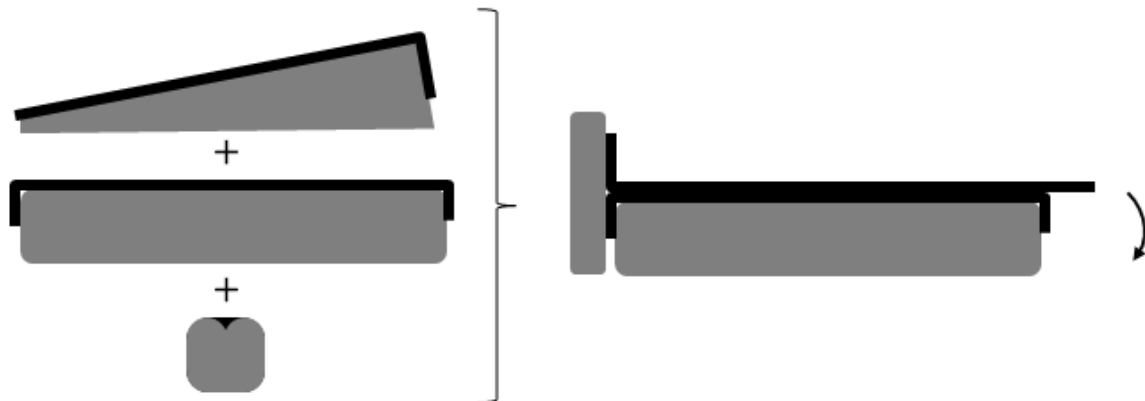


Figure 8 Spar J shape hot forming scheme

- **Infusion:** after placing the ancillary materials, the vacuum bag and the caul plates in their correct sites, the part is infused. The resin must be tempered and a leak test must be done before infusing.



Figure 9 Ancillary materials in a demonstrator before infusing (from Ref. 17)



Figure 10 Demonstrator after cured (from Ref. 17)

- **Trimming and drilling:** after infusion and demoulding the part, it is trimmed to get the final shape and the corresponding drill holes are made by numerical control machines.

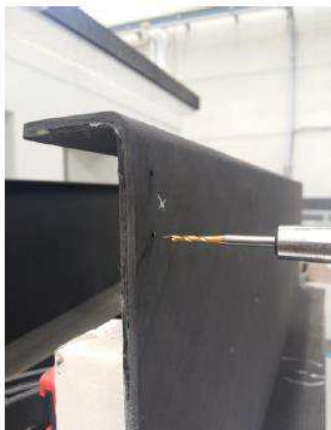


Figure 11 Drilling rear spar of demonstrator (from Ref. 17)

It is important to take into account the accessibility of the drilling head machine during the design of the parts.



Figure 12 Drilling head machine (from Ref. 17)

- **NDT inspection:** the technology used is laser-ultrasonic, which can be seen as an evolution of the conventional pulse-echo. The advantage of these inspections is that it is performed at a distance and without contact with the part, so there is no need for coupling agents and post-cleaning.



Figure 13 NDT inspection (from Ref. 17)

2.5 Process special considerations

-AFP special considerations:

- Minimum cut length (MCL): is the minimum length an AFP machine can cut and lay up. It depends on the AFP machine. Due to the MCL of the AFP machine (in this specific case is 135 mm), when dropping plies of $\pm 45^\circ$ some excess of material is placed.

This should be taken into account during the design of the plies, because some special shapes should be considered. In this specific case for pad-ups the next scheme has been considered:

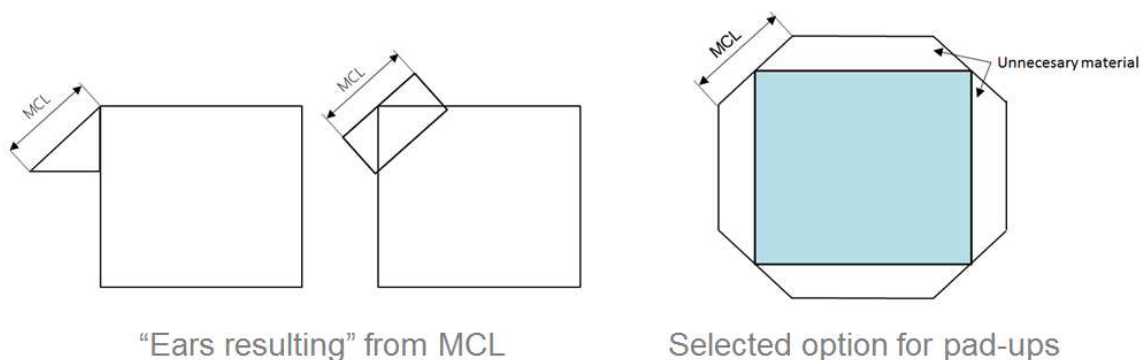


Figure 14 Shapes resulting from MCL

Specific details will be explained in following sections.

- Minimum concave radius: if it is laminated in curved tooling there is a limitation in feasible concave radius to lay up due to the geometry of the roller.

- Plies cutting tolerances: $\pm 0.2\text{mm}$

- Plies positioning tolerance (X, Y): $\pm 0.2\text{mm}$

- Pressure and temperature applied: they are specific for each material. For example, in this case a temperature of 170° is needed in order to melt the thermoplastic veil that is in the reinforcement (it works as a binder).

- Saerfix (ABS5931): is an epoxy resin binder film used for the first ply of the laminate. It is needed to stick the first ply on the AFP tooling. Although Manufacturing Engineering does not recommend its use (because it decreases the permeability of the laminate), during the test specimens manufacturing it has been considered necessary.

-Hot-forming special considerations:

- Spring back: it is important to take into account the effects of spring back during the manufacturing process and to try to minimize its effects, for example, with the design of the toolings.

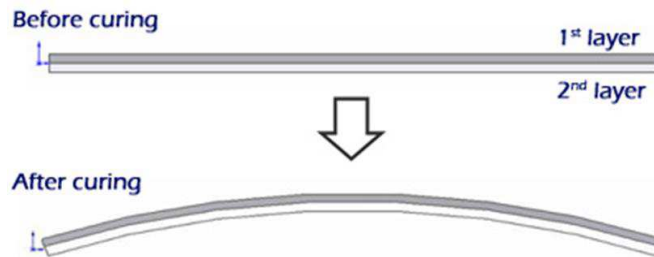


Figure 15 Schematics of spring-back

-LRI special considerations:

·Process duration and temperatures: for injection the resin must be preheat at 80°C, the mould at 120°C and the resin must be injected at 80°C und vacuum pressure from 1 to 5 bars

Standard Cure and Post-cure Cycle

Recommended cure cycle : 120 min at 180 °C - no postcure

Cure Cycle Possibilities

	Cycle N°1 (*)	Cycle N°2 (*)
Temperature		
180 °C	120 min	90 min
DMA dry - Glass Transition		
- E' onset value	202 °C	194 °C
- E'' peak	210 °C	206 °C
- Tan delta peak	215 °C	211 °C
Moisture weight gain (70°C/85%RH, equil)	0.80 %	0.84 %
DMA wet - Glass Transition		
- E' onset value	160 °C	155 °C
- E'' peak	170 °C	175 °C
- Tan delta peak	175 °C	190 °C

(*) Laminat Data G0926 SD (AS4C-J) E01 RTM6

Table 3 Hexcel RTM6 cure cycle

·Infusion techniques: apart from duration and temperatures, there are different infusion techniques which can be used. The quantity of resin in relation to fibre can be calculated. This was the technique used for coupons. However, in parts with complex shapes, sometimes more resin has to be infused in order to ensure that the resin reaches all areas of the part. Infusing more resin than needed could be one of the causes of overthickness, which is going to be studied in following sections.

Different techniques have been tested to try to minimize this effect. One manufacturability test was made using the technique of differential pressures. The deviations in thickness measured in this specimen were remarkably lower (only in certain areas), but zones with high porosity were found.

·Tooling: metallic male tools (thermal expansion coefficient has to be considered to design these toolings). Composite tools can be also used. It is important to assure that the tooling is correctly cleaned with MEK (Z-23.177) and to apply the demoulding agent

·Ancillary materials: LRI also needs some ancillary materials, such as:

- Vacuum bag film

- Release film and peel ply
- Resin media net
- Resin flow channel
- Sealant tape

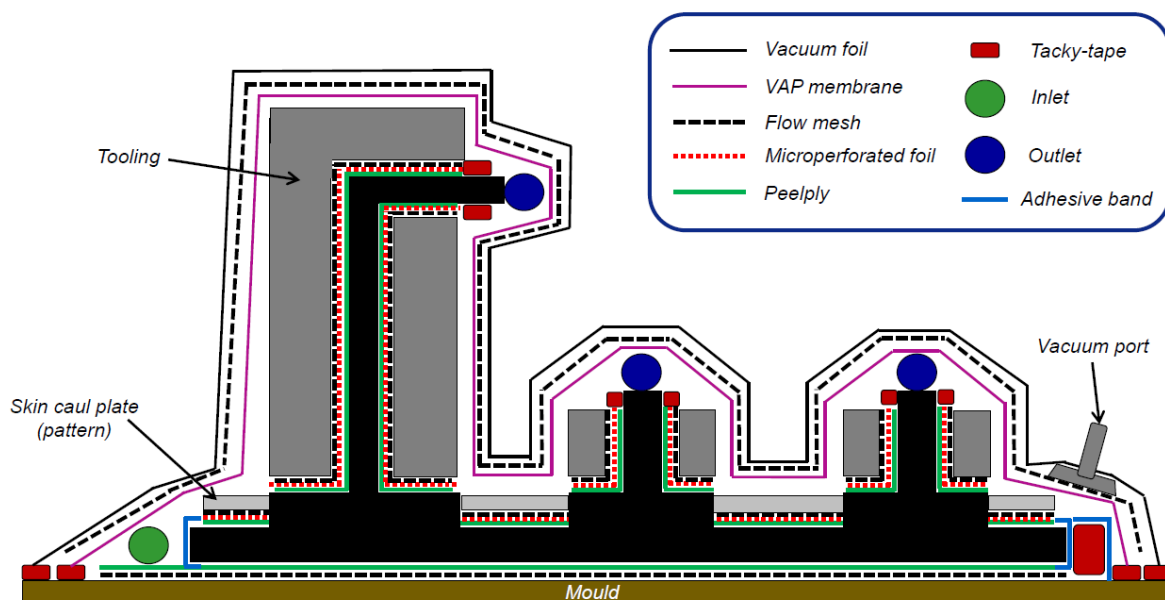


Figure 16 LRI Ancillary materials scheme

During the CS2 project it has been proved that the correct use and distribution of these ancillary materials have an important impact on the final characteristics of the parts.

3. DESIGN PRINCIPLES

3.1 General design principles

3.1.1 Rules for laminates

3.1.1.1 General Stacking Sequence Rules

For parts manufactured with LRI the general stacking sequence rules are also applicable (see Ref. 9). It is important to take into account that some rules go against others, so the use of one or another depends on the thickness, geometry, stress requirements...checking with Stress Department is always advisable.

The rules are the following:

- Symmetry: the laminate must be symmetric in respect to the mid axis. For each ply in $+\theta_i$ direction must be a ply in $-\theta_i$ direction in the symmetric position.
- Balanced: the laminate must be balanced. For each ply in $+\theta_i$ direction should be another ply of $-\theta_i$ direction.
- Plies orientation percentage: the percentage of plies in each direction depends on the element and its structural requirements. However there is a minimum percentage per orientation of 8% and a maximum of 67%.
- External plies: the external plies should never be in the direction of the main load. They should be a pair of $\pm 45^\circ$.
- Regular distribution: the plies with the same direction should be distributed along the stacking sequence to obtain a homogeneous stress distribution and to minimize coupling effects.
- Maximum grouping: the maximum number of plies placed together in the same direction is limited. The maximum number is four plies or one mm thickness.
- To improve buckling behavior: for compressive loads the 0° direction plies should be as far as from the symmetry line as possible
- To minimize coupling effects: it is recommended to group the $+45^\circ$ and -45° plies in pairs.
- To minimize inter-laminar shear effects: it is recommended to lessen the angle between adjacent plies.

3.1.1.2 Ply drop offs

-Slope: generally, in thicknesses changes the maximum slopes allowed are:

- 1/20 for load direction
- 1/10 for other directions

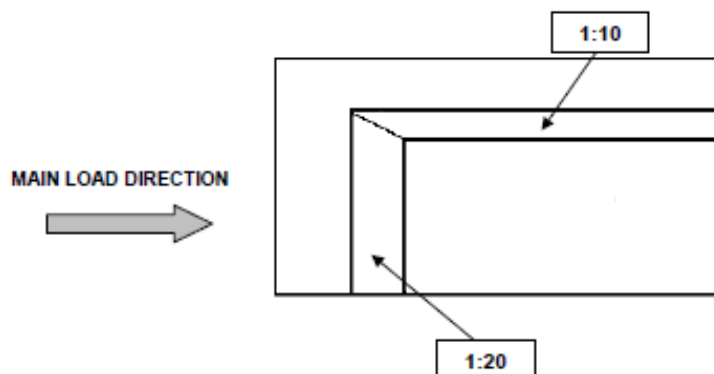


Figure 17 Maximum slopes allowed

Particular cases will be described in subsequent sections of this document.

- Staggering patterns: to optimize structural behavior the plies should drop the closest to the symmetric axis.
- External plies: the two most external plies should be continuous
- Covering plies: in general, every four dropped plies there should be one ply covering those dropped plies.

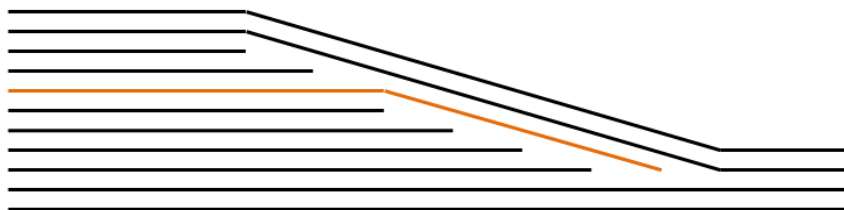


Figure 18 General case for covering plies

However for this specific case, due to a peculiarity of the AFP machine used for the test specimens manufacturing, there should be one ply covering those dropped plies every two dropped plies. This condition must be reviewed if the AFP machine is changed.

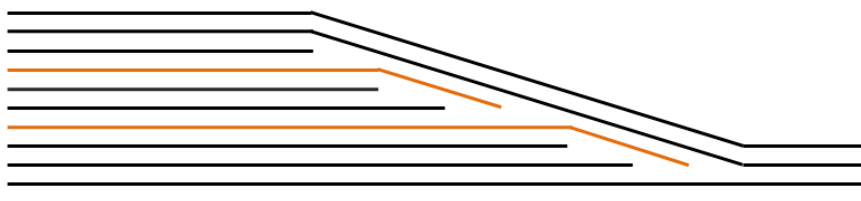


Figure 19 Specific case for covering plies in LRI

- Dropping plies at the same position: avoid dropping adjacent plies at the same point. If it is necessary, the distance should be the biggest possible.
- Uniform distribution: the dropping plies should be distributed along the laminate
- Symmetry and balance: it is recommended to keep symmetrical and balanced the drop-off areas

3.1.1.3 Ply splices

-Splice line parallel to fibre direction: overlaps are not allowed, because they do not produce any added value and they cause over-thickness. The gaps between tapes should be between 0 and 2,5 mm.

Avoid splicing plies at the same position using a stagger of 21 mm. There should be at least 4 plies between coincident gaps.

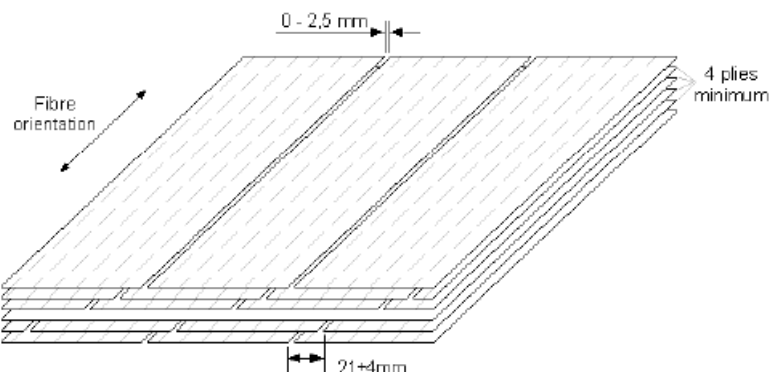


Figure 20 Splice line parallel to fibre direction (from Ref. 9)

-Splice line perpendicular to fibre direction: gaps are not allowed, because it would mean loss of continuity. Overlaps are not preferred but, if necessary, it will be 12-25 mm overlap. For adjacent plies, there should be at least 2000 mm between two joints.

Overlaps should be avoided in areas where the thickness is sensitive to the assembly process, due to the over-thickness it causes.

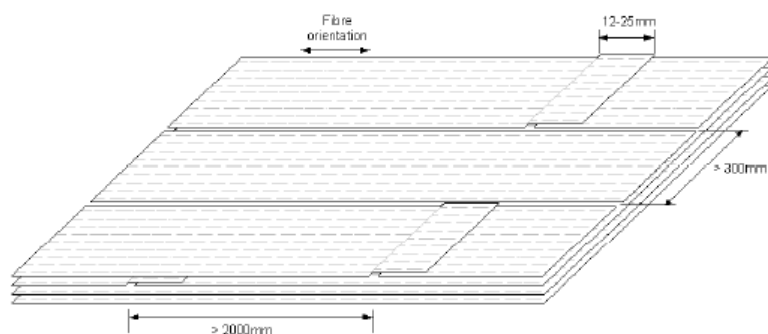


Figure 21 Splice line perpendicular to fibre direction (from Ref. 9)

3.1.2 Surface contour general tolerances

In Ref. 4 some standard values are specified:

- For non-aerodynamic surfaces the surface contour should have a tolerance of ± 0.75 mm.
- For aerodynamic surfaces the surface contour should have a tolerance of ± 0.50 mm.
- For joint surfaces the surface contour should have a tolerance of ± 0.50 mm.

These values may change depending on the specific characteristics of the wing box structure components. If that is the case, they will be defined in the corresponding sections of this document.

3.1.3 Bend radii

Internal radii should be 3 mm minimum. Furthermore they should measure at least the value of the thickness.

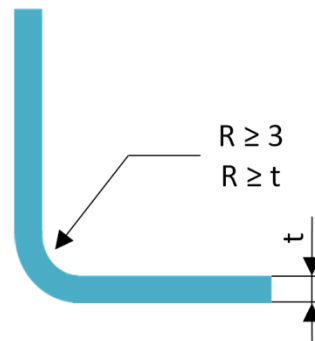


Figure 22 Internal minimum radii

The tolerance of these radii is $\pm 0,5$ mm.

3.1.4 Joints

One of the main purposes of LRI manufacturing is to achieve high structural integration by infusing “one shot components” and to avoid any kind of joints. However, the infused part must be connected to the rest of the structure, in which case bolted joints shall be used.

3.1.4.1 Fasteners distances

The typical fasteners distances are:

	Edge dist. (Norm. to load)	Edge dist. (Load dir)	Pitch
Fastener Ø	2,5Ø+1	3Ø+1	4,5Ø+1
4 / 4,2	11,5 ⁽¹⁾	13,5 ⁽¹⁾	20 ⁽¹⁾
4,8	13	15,5	22,5
6,4	17	20	30
7,9	21	25	37

- (1) Due to reparability oversize availability, use Ø4.8 values for Hi-lite, Hi-lok, Lockbolts and blind fasteners. In this case, for countersunk fasteners, minimum skin thickness must be checked in order to keep the minimum cylindrical area.

Table 4 Fastener distances

3.1.4.2 Fastener selection

For rib union in test specimens, DAN6 hi-lok and NSA5075 nut have been used. This hi-lok head is smaller, so it is recommendable for thinner thickness (in order to avoid sharp edges, the thickness of a fastener should be maximum $\frac{2}{3}$ of the thickness of the part in which the countersunk is made).

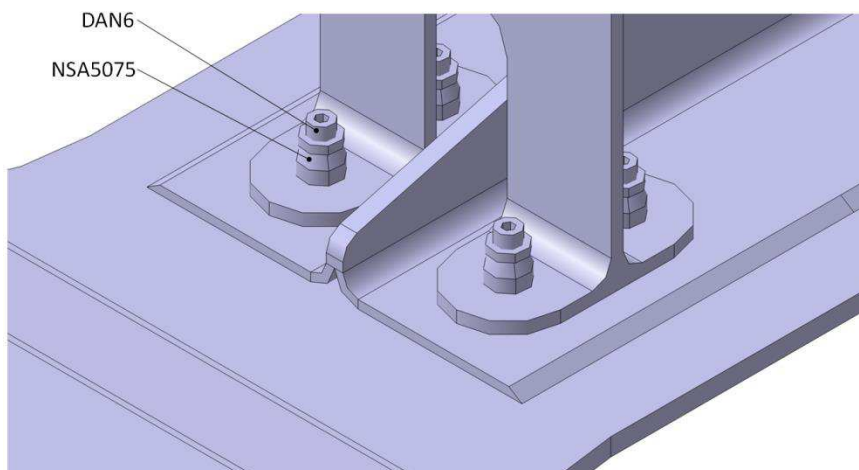


Figure 23 Fastener selection

Hi-lites can be also used for thicker thickness, in which the laminate thickness allows for its countersunk head shape, taller than for hi-locks

In addition, protuberant head hi-lites can be used for non-aerodynamic surfaces joints, such as joints to spar web.

3.1.4.3 Distances to ramp edges

There should be a minimum of 5 mm distance from longitudinal elements to drop-off edges to cover manufacturing and positioning tolerances.



Figure 24 Distance to ramp edges

This minimum distance is influenced by:

- Stringer foot width tolerance
- Stringer positioning tolerance

-Ramp positioning tolerance (tow positioning and cutting tolerance of the machine)

3.1.4.4 Drill break-out protections

Usually an external GFRP ply is needed in order to protect composite parts against delamination during the drilling process.

For test specimens, this protection was not deemed necessary due to the drilling process characteristics, in which either sacrificial material was used or the holes were transferred from jig jaws.

For in-flight demonstrator may be possible to use these protections.

Drill break-out of LRI composite parts in comparison with typical prepreg composites: TBD

3.1.5 Corrosion Protections

The corrosion protection schemes used are according to Ref. 6 and based in the wing susceptibility zones defined in Ref. 16. Non-metallic components are intrinsically corrosion resistant and in general do not require any specific protection.

However, specific means of protection has to be taken for assembly with metallic parts. In this case an external ply of GFRP has been used to protect CFRP parts. In test specimens it was infused with the rest of the laminate.

The GFRP ply should cover the interface with metallic parts as shown in the following figure:

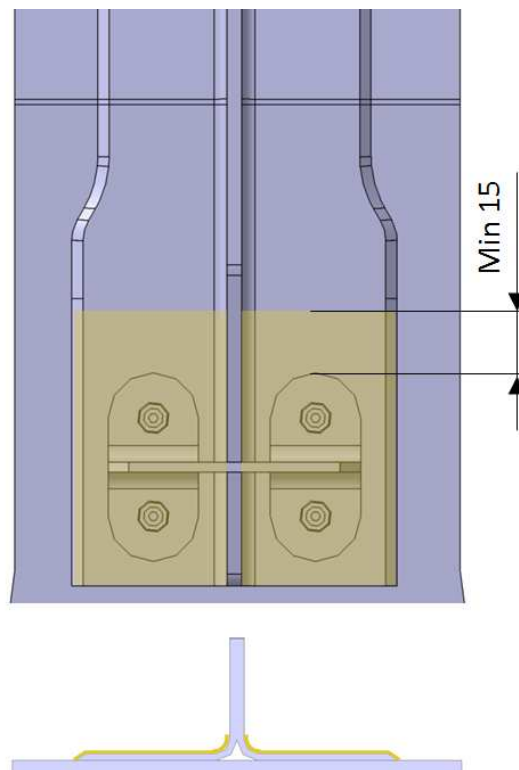


Figure 25 Corrosion protections

3.1.6 Lighting strike protection

LRI is a new manufacturing process for primary structures so it is mandatory to study its behavior against lighting strike. For that reason a test plan for simulated lighting strike tests has been made, which is explained in Ref. 7.

The Outer External Wing Box is located mainly in Zone 3 except a Zone 2A, 500mm wide, in the outer face of the covers. For zone 3 no additional protections are needed, taking into account that the thicknesses are higher than 2 mm and there is no fuel in the outer external wing box.

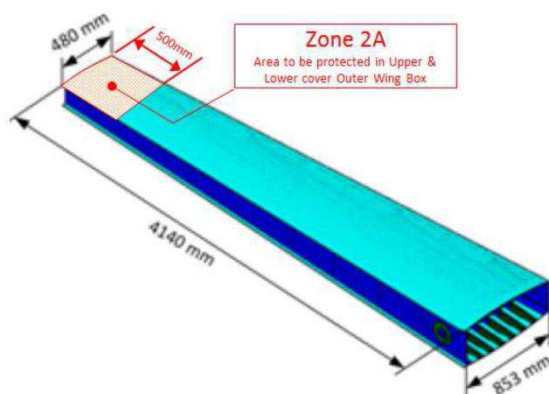


Figure 26 CS2 Outer external wing zoning (from Ref. 7)

The protection considered for zone 2A is a bronze mesh. One lighting strike protection requirement is not to leave the mesh exposed. Bronze mesh integration tests have been made to study the possibility of including the mesh in the laminate and infusing, to integrate it completely.

3.1.7 Fire Worthiness Requirements

In this case, special fire worthiness requirements have not been considered.

3.1.8 Reparability

Reparability is one of the main design drivers for composite parts. It is important to take into account reparability during the design phase, for example, to size the structure to incorporate a bolted repair, as it would be explained in following sections.

3.1.9 Manufacturing tolerances

Manufacturing tolerances are specified in the section 4 of this document (Dimensional study of test specimens). For those which are not included, Ref. 8 must be addressed. They could be reconsidered during the following design steps.

3.2 Specific design features

3.2.1 General Description

The outer external wing box lower skin structural concept is a one shot skin with integrated rear and front spars and T shape stiffeners. The overall part dimensions are 4128x980x330 mm (LxWxH). Based on this design, specific design features are being explained for these three types of components:

- Stringers (T shape)
- Spars
- Skin

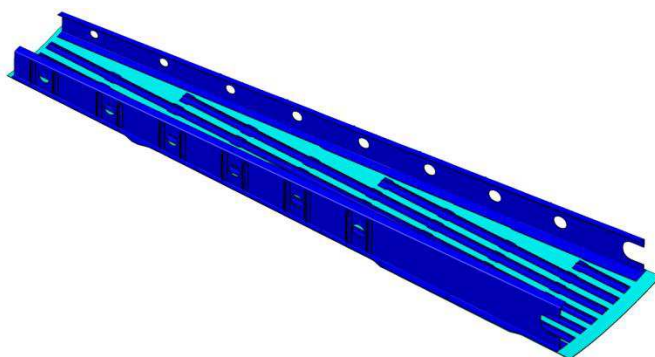


Figure 27 Outer external wing box lower skin

The rear spar is taken as datum for tolerance positioning. This is the scheme:

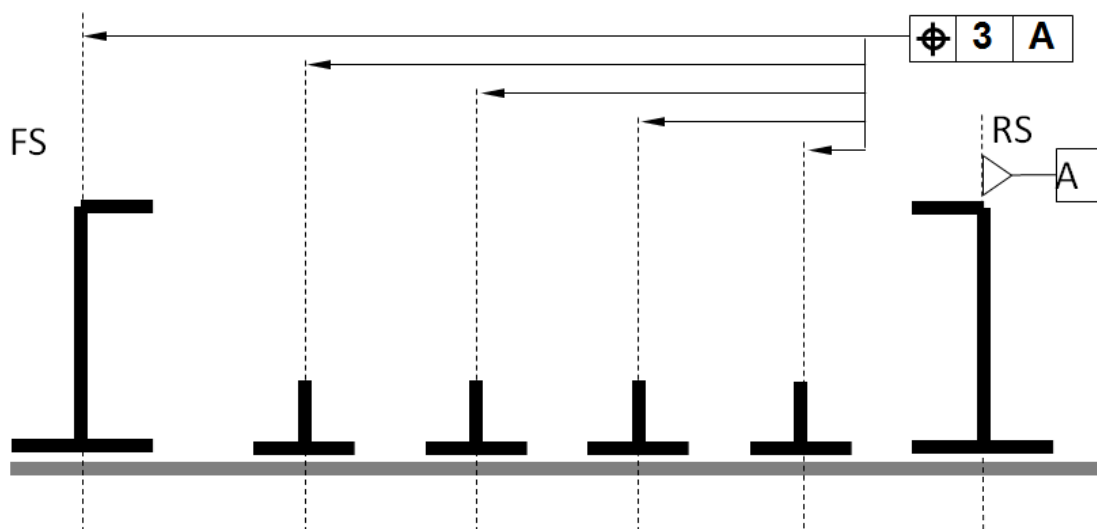


Figure 28 Tolerance positioning criteria

In the following sections, different dimensions and design features are expressed as a function of other parameters, such as fastener diameter and component thickness. When these are replaced by numerical values, it must be considered that they could vary in subsequent design steps, so they can only be taken as a reference.

3.2.2 Stringer

3.2.2.1 Stringers concept

Stringers designed for this external wing box have a T-shape and are obtained from two L-shapes parts



Figure 29 T-shape stringer

The free space under the stringer between the two shapes will be filled with the roving. The shape of this roving is determined by the thickness of the L-shapes. As the external radii of this parts is constant, the roving can be manufactured from a constant rod conformed with the proper shape.

3.2.2.2 Stringer minimum radii

The minimum internal stringer radii should be 3 mm. In addition, it should be at least the value of the thickness. External radii are set constant, so the roving shape can be constant along the span

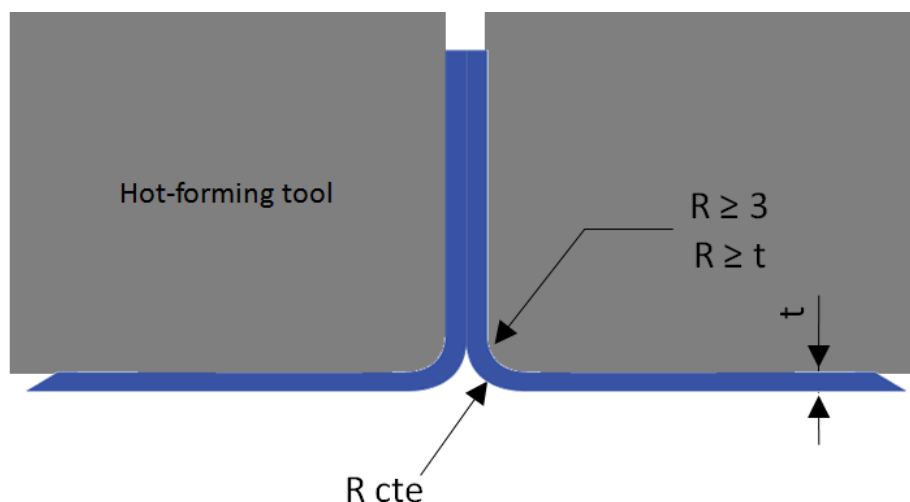


Figure 30 Stringer minimum radii

The radii tolerance agreed with manufacturing is $\pm 0,5\text{mm}$.

3.2.2.3 Stringer foot width

Although the stringer is cocured to the skin, the minimum foot width is limited by a possible refurbishment with a riveted machined profile. In this case is affected by:

- Web thickness
- Minimum internal radius

- Machined profile thickness
- Tool machine radius
- Tolerance of trimming
- Fastener diameter (and its corresponding typical distances)

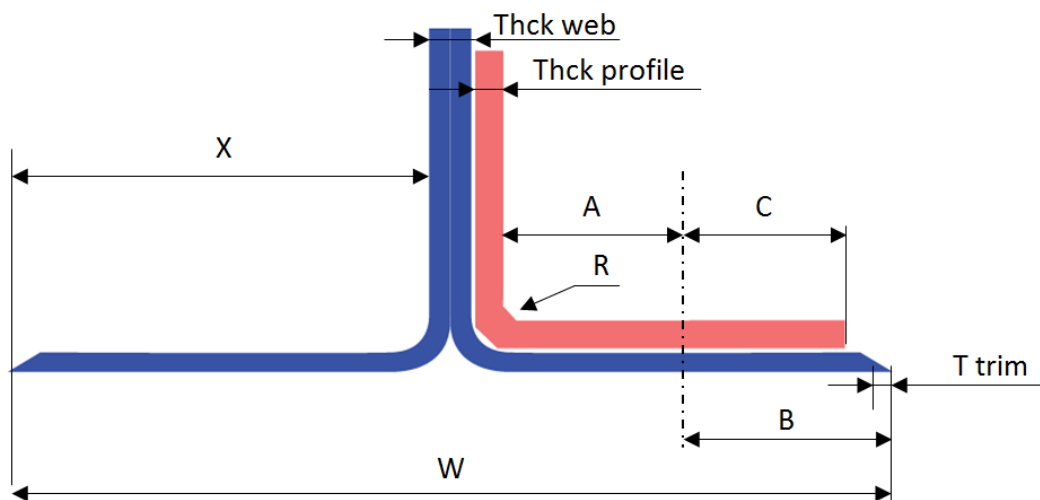


Figure 31 Stringer foot width scheme

For standard stringer section of web thickness equal to 3,5 mm, profile thickness of 2,5 mm and trimming tolerance of 0,75 mm:

	STRINGER FOOT WIDTH				
	A	B	C	X	W
ø fastener	$\phi+1+R$ (min 11)	$2,5\phi+1+Ttrim$	$2\phi+1$	$A+B+Thck\ profile$	$2X + Thck\ web$
4,2	11	12,25	9,4	25,75	55
4,8	11	13,75	10,6	27,25	58

Table 5 Stringer foot width distances

The distances given in the tables do not consider the bevel. When considering it, an additional distance per bevel must be added (additional distance per bevel= thickness/tag32.5°)

3.2.2.4 Stringer foot width on rib interface

The minimum foot width in rib zone is limited by the riveted union to the rib. In this case is affected by:

- Stringer web thickness
- Stringer internal radius
- Stringer positional tolerance
- Detail design of rib foot

- Tolerance of trimming
- Fastener diameter (and its corresponding typical distances)

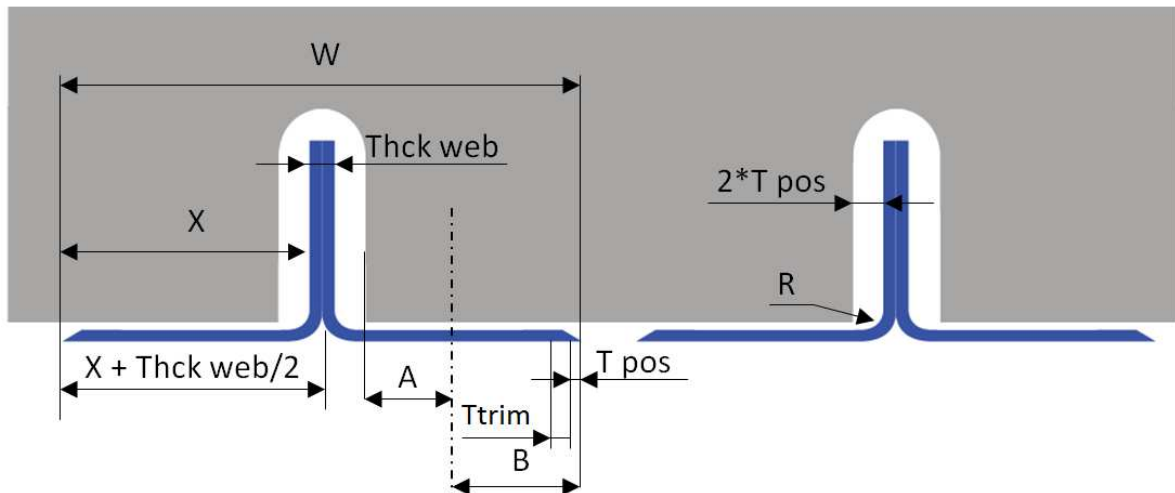


Figure 32 Stringer foot width on rib interface

For standard stringer section of web thickness equal to 3,5 mm, positioning tolerance of 1,5 mm and trimming tolerance of 0,75 mm:

	STRINGER FOOT WIDTH ON RIB INTERFACE			
	A	B	X	W
ø fastener	$2\phi+1$	$2,5\phi+1+T_{trim}+T_{pos}$	$A+B+2*T_{pos}+1+R$	$2X + Thck\ web$
4,2	9,4	13,75	30,15	64
4,8	10,6	14,25	31,85	67
6,4	13,8	18,25	39,05	82

Table 6 Stringer foot width on rib interface distances

The distances given in the tables do not consider the bevel. When considering it, an additional distance per bevel must be added (additional distance per bevel= thickness/tag32.5°)

3.2.2.5 Stringer foot profile on rib interface

The length of the stringer foot contour for rib union will vary with the shape of the rib (T or C). Is affected by:

- Rib thickness
- Fastener diameter (and its corresponding typical distances: distance from fastener to rib, distance from fastener to edge)

Ribs are joined only to stringers (not to the skin).

The shape of the rib may impact these dimensions. For example, the need of a vertical stiffener close to the 'mouse hole' will have an impact on the rivet distance to rib web, hence increasing the profile overall length

For T shape rib, the scheme is the following:

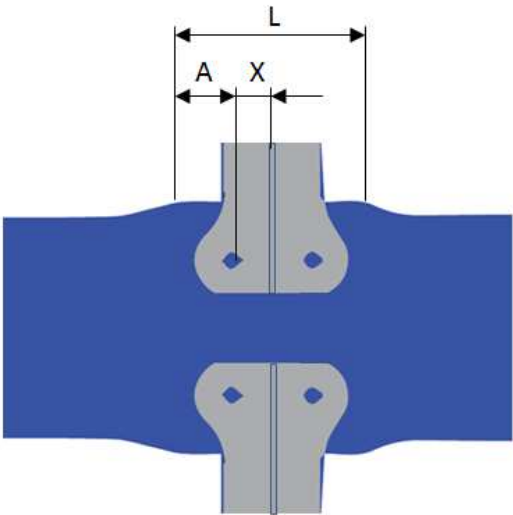


Figure 33 Stringer foot profile on rib interface in T scheme

For rib web thickness equal to 5 mm:

	STRINGER FOOT PROFILE ON RIB INTERFACE (T SHAPE)		
	A	X	L
ø fastener	$3\phi+1$	$\phi+1+R$	$2X + 2A+ RibwebThck$
4,2	13,6	11	41
4,8	15,4	11	45
6,4	20,2	12,4	58

Table 7 Stringer foot profile on rib interface in T scheme distances

For C shape rib, the scheme is the following:

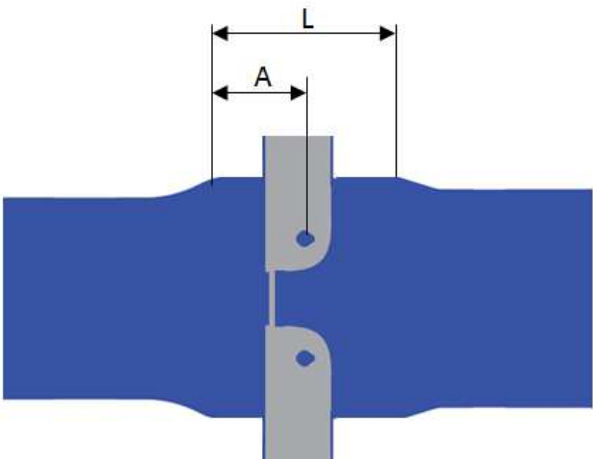


Figure 34 Stringer foot profile on rib union in C scheme

	STRINGER FOOT PROFILE ON RIB INTERFACE (C SHAPE)	
	A	L
ø fastener	3ø+1	2A
4,2	15,4	31
4,8	15,4	31
6,4	20,2	40

Table 8 Stringer foot profile on rib union in C scheme

For ø4.2 fasteners the ø4,8 values are considered in order to achieve a minimum distance for rib attachment.

3.2.2.6 Stringer height

The maximum stringer height is limited by the thickness of the web, in order to avoid stringers web buckling. The height should be equal or lower than ten times the web thickness.

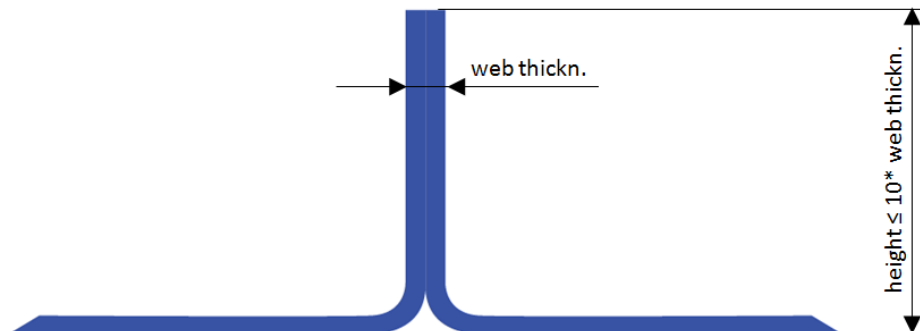


Figure 35 Stringer height

3.2.2.7 Stringer run-out detail

The stringer run-out dimensions are limited by rib union. It is influenced by:

- Rib thickness
- Fastener diameter (and its corresponding typical distances)

For T shape rib, the scheme is the following:

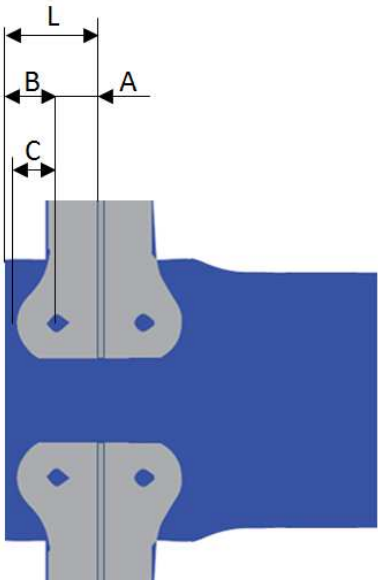


Figure 36 Stringer run-out rib union in T scheme

For rib web thickness equal to 5 mm and radius 4 mm:

	STRINGER RUN-OUT DETAIL (T SHAPE)			
	A	B	C	L
Ø fastener	$\phi+1+R$ (min 11)	$3\phi+1$	$2\phi+1$	$A+B+RibwebThck/2$
4,2	11	13,6	9,4	27
4,8	11	15,4	10,6	29
6,4	11,4	20,2	13,8	34

Table 9 Stringer run-out detail (T shape) distances

The end of the stringer run-out is also related to the end of the foot enlargement, and must accomplish a minimum of 10 mm (the stringer run out finishes 10mm before than the foot enlargement): This is a Stress requirement, and it is applicable for T and C ribs.

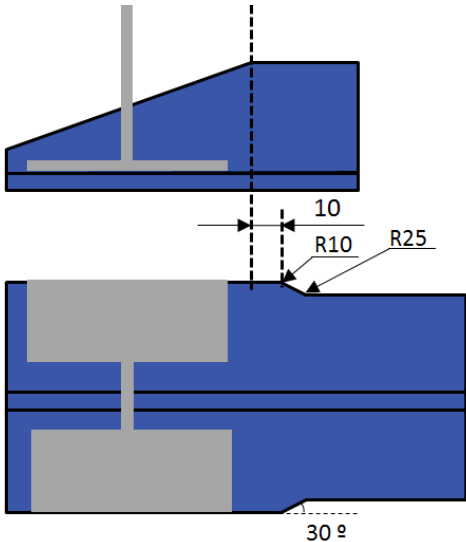


Figure 37 Stringer run out relation with foot enlargement

For C shape rib, the scheme is the following:

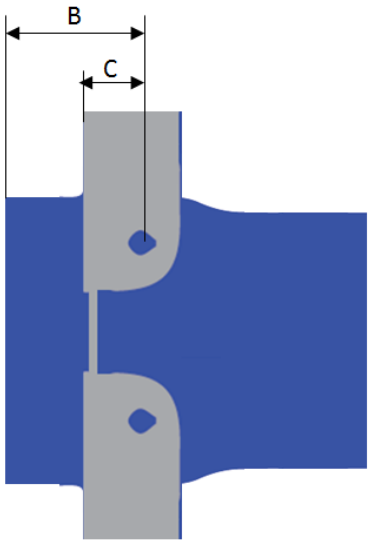


Figure 38 Stringer run-out rib union in C scheme

For radius equal to 4 mm:

	STRINGER RUN-OUT DETAIL (C SHAPE)	
	B	C
ø fastener	$3\phi+1$	$\phi+1+R$ (min 11)
4,2	13,6	11
4,8	15,4	11
6,4	20,2	11,4

Table 10 Stringer run-out detail (C shape) distances

3.2.2.8 Stringer height change detail

The stringer radius and angles for height change considered are:

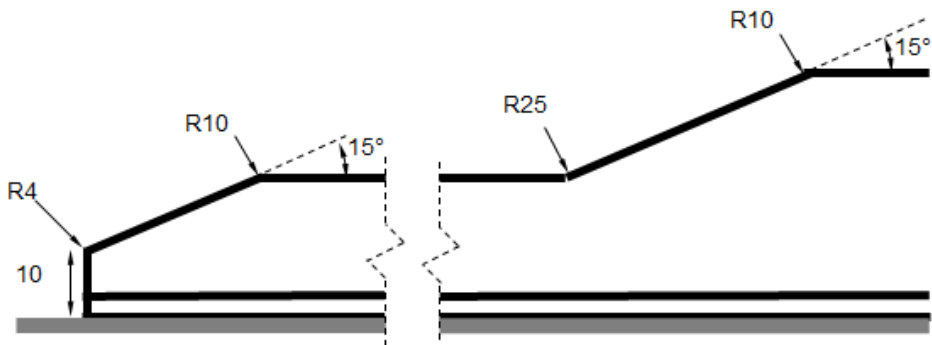


Figure 39 Stringer height change detail (from Ref. 10)

3.2.2.9 Stringer section and thickness detail

The minimum distances stringer details between stringer geometrical changes are:

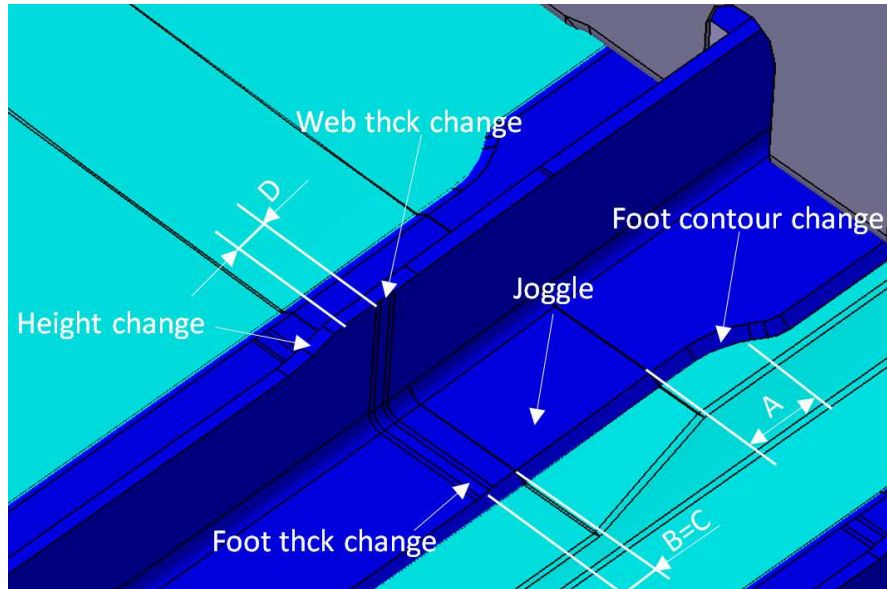


Figure 40 Stringer section and thickness detail

A=Minimum distance between stringer foot contour change to skin thickness change=10mm

B=Minimum distance between skin thickness change to stringer foot thickness change=5mm

C= Minimum distance between foot thickness change to stringer web thickness change =5mm

D=Minimum distance between stringer web thickness change to stringer height change= 10mm

The sequence in which these geometrical changes occur is the one shown in the previous figure and was agreed with Stress and Manufacturing Department.

3.2.2.10 Stringer pitch

The pitch between stringers is influenced by stress requirements (see Ref. 2). The value given by Stress requirements in Ref. 2 is 110 mm.

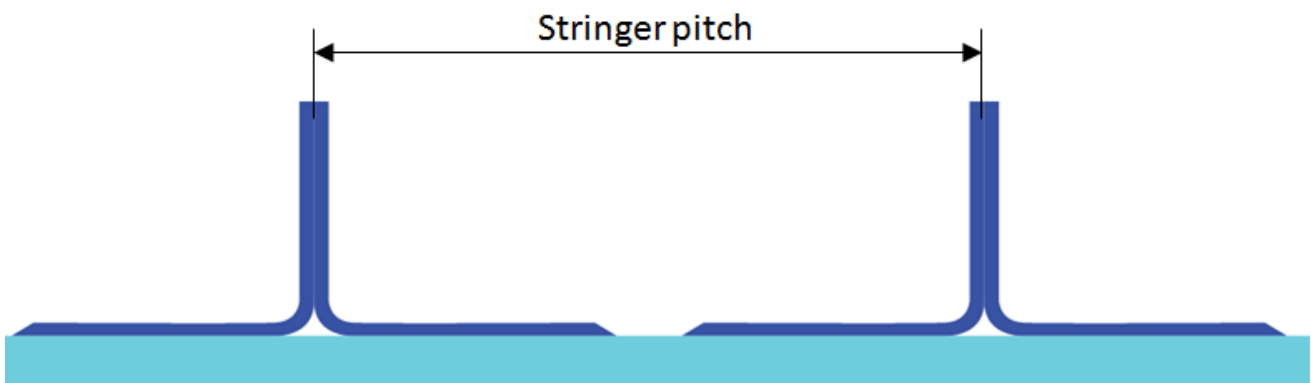


Figure 41 Stringer pitch scheme

3.2.2.11 Stringer feet bevels

Due to the manufacturing process, when the T shapes of the stringers are bent, an external bevel in the foot of the stringers is created (known as “book effect”). It has to be considered when calculating the elements final dimensions and tolerances.



Figure 42 Stringer feet bevels

The angle of the bevel agreed with manufacturing is 32.5°. This angle comes from the following scheme:

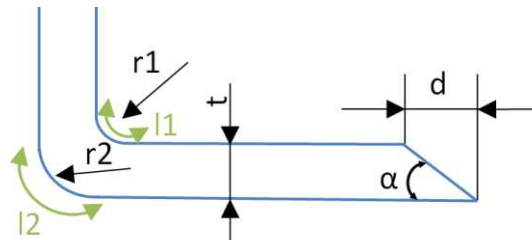


Figure 43 Stringer feet bevel scheme

$$d = l1 - l2 = (\pi/2) * (r1 - r2) = (\pi/2) * t$$

$$\tan \alpha = t/d = t/((\pi/2) * t) = 2/\pi$$

Then the angle of the bevel is independent of the thickness and it has a value of:

$$\alpha = \arctan(2/\pi) = 32.48^\circ \approx 32.5^\circ$$

3.2.2.12 Stringer stacking sequence particularities

As it has been explained before, the stringers considered are obtained from two L-shapes parts. The design solution used for the laminate is an anti-symmetrical lay-up for stringer web.

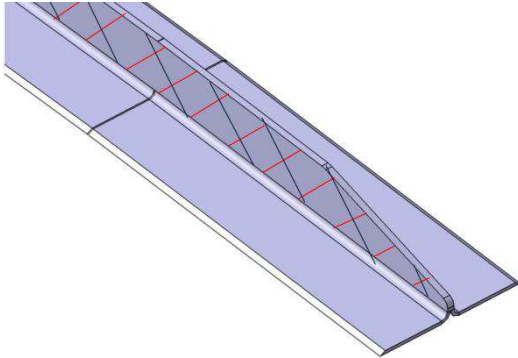


Figure 44 Stringer web stacking sequence detail

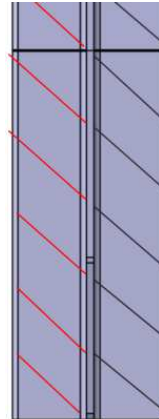


Figure 45 Stringer feet stacking sequence detail

Therefore, the fibre orientations in stringer feet are the same. The external ply orientation of the feet stringer in contact to the skin should be the same that the first ply of the skin.

3.2.3 Spar

3.2.3.1 Spars concept



Initially, front spar was an H shape, created from two C shapes and with a cap added on the upper feet to obtain the same laminate that in web section. Finally the H shape was not considered necessary and J shape was adopted for both rear and front spars.

The J shape is obtained from a C shape and a Z shape and a roving.

Figure 46 Spars concept

J-shape was necessary to have a double-flanged union with the lower skin, as the union is cocured and the symmetry helps to stand peel stresses. The interface with upper skin is bolted, and C-shape union has been considered appropriate.

During the design phase three different concepts of spar have been studied: the first with flat web outer surface flat, the second with flat web inner surface and the third with symmetric web.

A comparing matrix was done including the implications from different points of view (design, stress, manufacturing...). Each implication was provided by the corresponding department.

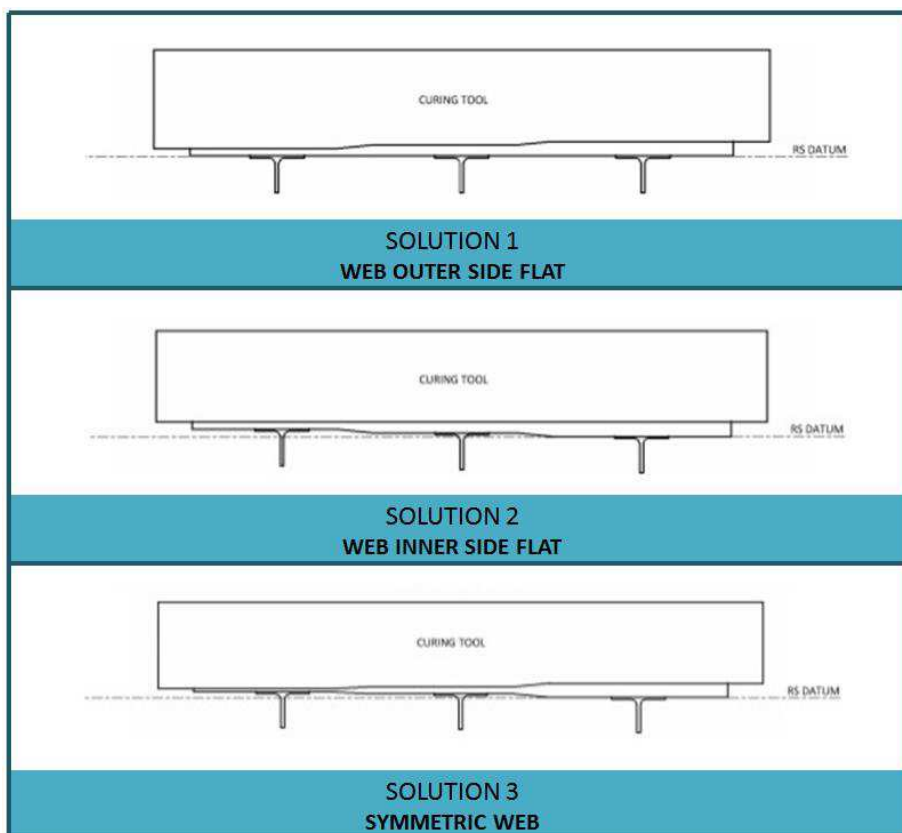


Figure 47 Different concepts of spar studied

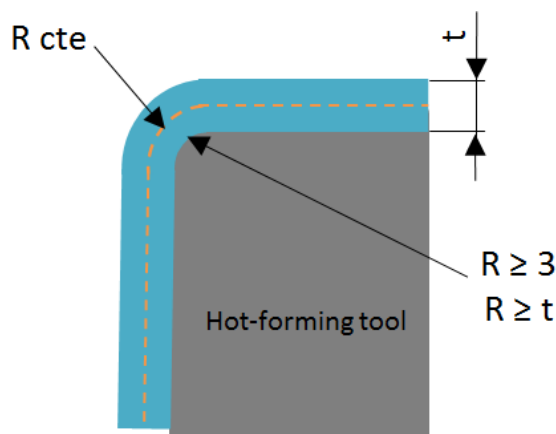


Figure 48 Spar minimum radii

The radii tolerance agreed with manufacturing is $\pm 0,5\text{mm}$.

3.2.3.3 Minimum spar cap width

The minimum width of internal (box side) spar caps is limited by different cases for lower or upper cap. In both cases is affected by:

- Web thickness
- Minimum internal radius
- Machined profile web thickness
- Tool machine radius
- Tolerance of trimming
- Fastener diameter (and its corresponding typical distances)

-Spar lower caps (T section): cocured union with lower skin. The width of spar lower cap is limited by stress requirements or by a possible refurbishment with a riveted machined profile with a single riveting line (the most restrictive condition). Scheme for minimum width for riveted machined profile is:

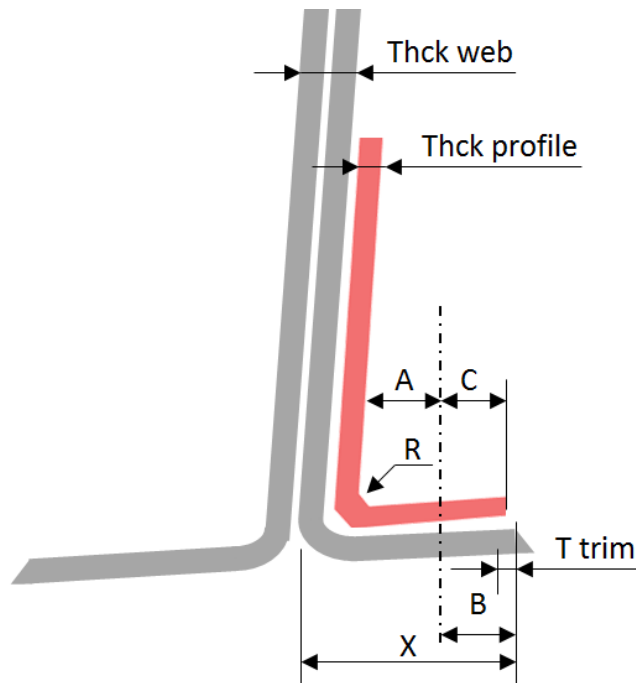


Figure 49 Spar inner lower cap scheme

For spar section of thickness equal to 5.1 mm (maximum value included in Ref. 3), profile thickness of 2,5 mm and trimming tolerance of 0,75 mm:

	SPAR LOWER CAP WIDTH			
	A	B	C	X
ø fastener	$\phi+1+R$ (min 11)	$2,5\phi+1+T_{trim}$	$2\phi+1$	$A+B+Thck\ profile+Thck\ web/2$
4,2	11	12,25	9,4	28
4,8	11	13,75	10,6	30
6,4	11,4	17,75	13,8	34

Table 12 Spar lower cap width distances

The distances given in the tables do not consider the bevel. When considering it, an additional distance per bevel must be added (additional distance per bevel= thickness/tag32.5°)

-Spar upper cap (L section): riveted union with upper skin. Riveting scheme is a double riveting line at 45°. The width of spar lower cap is driven by riveting joint lay out, but making it possible to add a riveted machined profile in case refurbishment is needed, using the existing rivet positions.

Scheme for minimum width for riveted machined profile considering two riveting lines is:

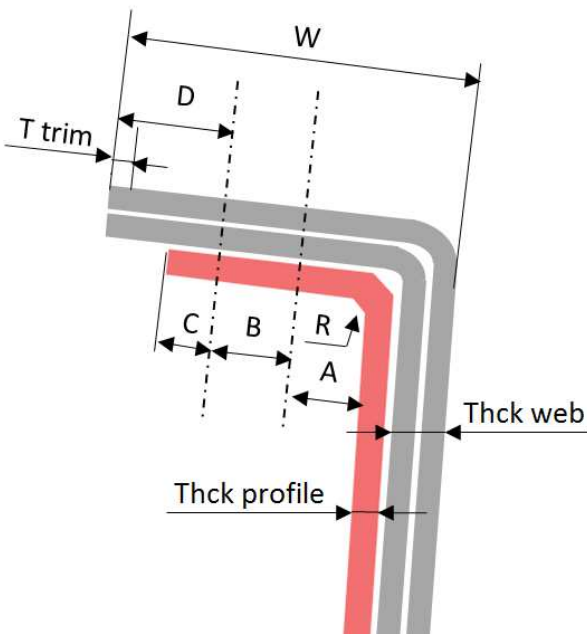


Figure 50 Spar upper cap scheme

For spar section of thickness equal to 5.1 mm (maximum value included in Ref. 3), profile thickness of 2,5 mm and trimming tolerance of 0,75 mm:

	SPAR UPPER CAP WIDTH				
	A	B	C	D	W
ø fastener	$\phi+1+R \text{ (min 11)}$	$(4,5\phi+1) \cdot \sin 45^\circ$	$2\phi+1$	$2,5 \phi+1 + T_{trim}$	$A+B+D + Thckweb+Thck \text{ profile}$
4,2	11	19,61	9,4	12,25	50
4,8	11	22,31	10,6	13,75	55
6,4	11,4	29,51	13,8	17,75	66

Table 13 Spar upper cap width distances

3.2.3.4 Minimum width of spar cap on rib interface

The minimum width of spar cap on rib interface is driven by the rib to spar flange joint, for which two parallel lines of riveted are deemed necessary. This statement is still to be confirmed by Stress Department

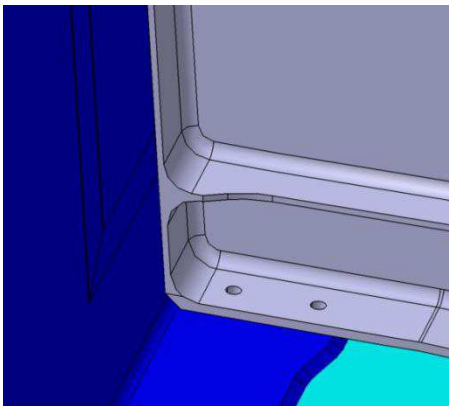


Figure 51 Spar cap on rib interface

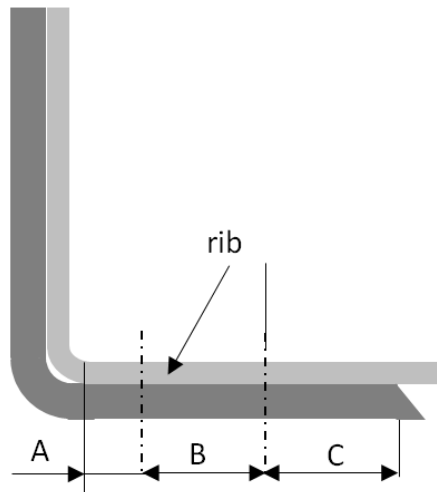


Figure 52 Spar cap on rib interface scheme

	SPAR CAP WIDTH ON RIB INTERFACE		
	A	B	C
ø fastener	$\phi+1$	$(4,5\phi+1) * \sin 45^\circ$	$3\phi+1$
4,2	5,2	14,13	13,6
4,8	5,8	16,05	15,4
6,4	7,4	21,16	20,2

Table 14 Spar cap on rib interface

The distances given in the tables do not consider the bevel. When considering it, an additional distance per bevel must be added (additional distance per bevel= thickness/tag32.5°)

3.2.3.5 Lightening holes

Lightening holes have been added to spar web in the zone between ribs where possible (see Ref. 3). The shape considered for them is circular and their centers are placed in the geometrical center of the trapezoid defined by rib planes and lower and upper skin.

The size of the holes diameter is the third part of the maximum height of the spar web following stress requirements explained in Ref. 3.

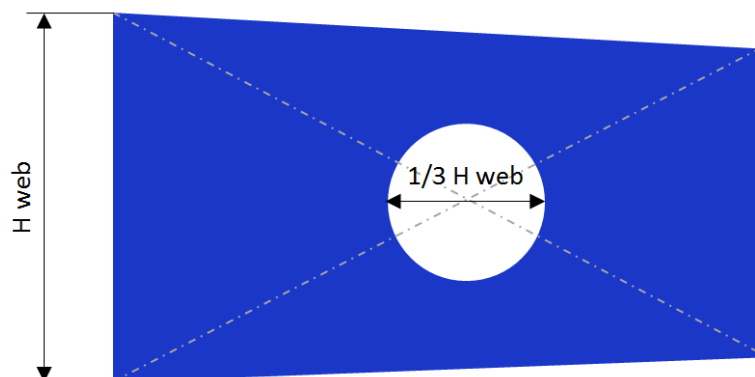


Figure 53 Lightening holes spar web

3.2.3.6 Spar stiffeners

To stabilize the spar webs, two stiffeners will be added to both sides of the lightening holes of spars where necessary due to stress requirements (see Ref. 3).

They are T-shape and their design features are similar to stringers previously shown.

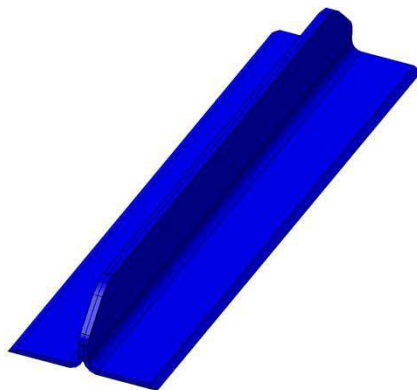


Figure 54 Spar stiffeners

They include the following run-out in both sides:

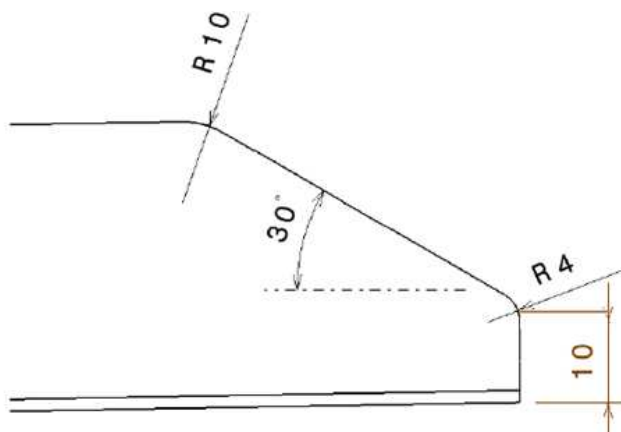


Figure 55 Spar stiffeners run-out detail

3.2.3.7 Stiffeners distribution

As it has been said before, stiffeners are added when necessary in accordance with stress requirements.

The distance from spar stiffener mid axis to the end of the rectangle between ribs should be 1/3 of the trapezoid width defined by rib planes and lower and upper skin width following stress requirements explained in Ref. 3.

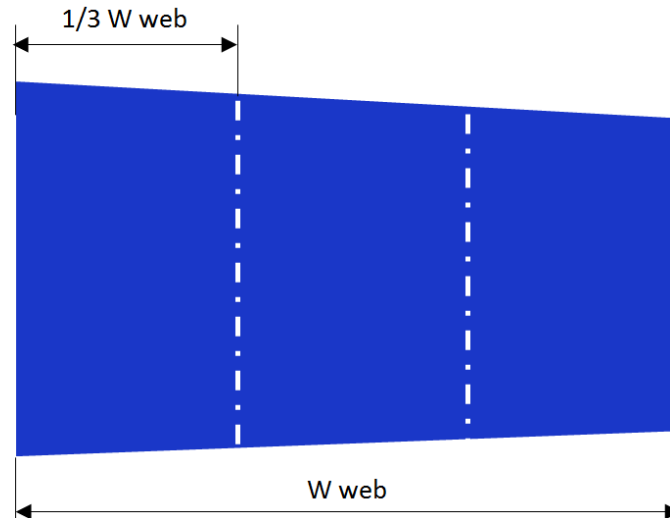


Figure 56 Spar stiffeners position

Both stiffeners should be equal, and its length is limited by a minimum distance of 5 mm from the end of the stiffener to the beginning of both spar caps radii and a maximum of 10 mm.

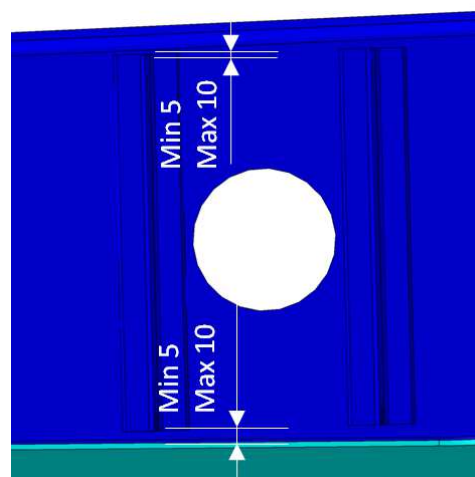


Figure 57 Spar stiffeners minimum and maximum distances

The minimum distance is limited by the stiffener closer to the wing tip and set to 5 mm, and the maximum by the stiffener closer to the root position.

3.2.3.8 Web local overthickness at rib interface

Local overthickness could be necessary to reinforce local zones in spar web on rib interfaces, although this requirement is still not included in preliminary sizing by Stress Department in Ref. 3

The width of the overthickness (W) depends on the orientation of the plies that should be added and the minimum cut length of the AFP machine. In this specific case where MCL is 135:

-For 0° direction plies: equal to MCL of the AFP machine

$$W=135 \text{ mm}$$

-For 90° direction plies: a multiple of tow width, ensuring a minimum distance of 10mm from rib flange to the end of the overthickness

$$W=n \cdot \text{tow width (12.70mm)}$$

-For +/- 45° direction plies: cosine of 45° times MCL

$$W= \cos 45^\circ \cdot 135 = 96 \rightarrow 100 \text{ mm}$$

In the next scheme it is illustrated the W mentioned before, considering t1 thicker than t2.

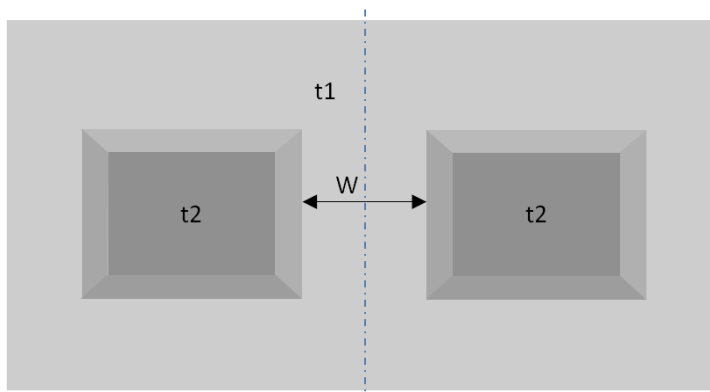


Figure 58 Web local overthickness at rib interface

In all cases a minimum distance of 10 mm from rib flange to the end of the overthickness is necessary.

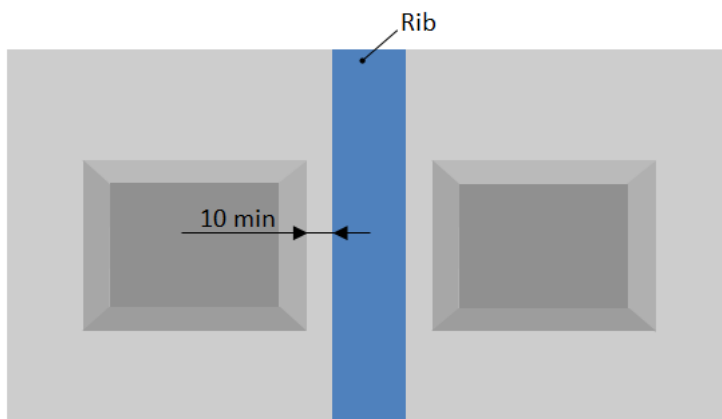


Figure 59 Minimum distance from rib flange to end of local overthickness

3.2.3.9 Spar thickness changes

Typical thickness changes related geometry for spars have been considered. The slopes values are shown in the next figure:

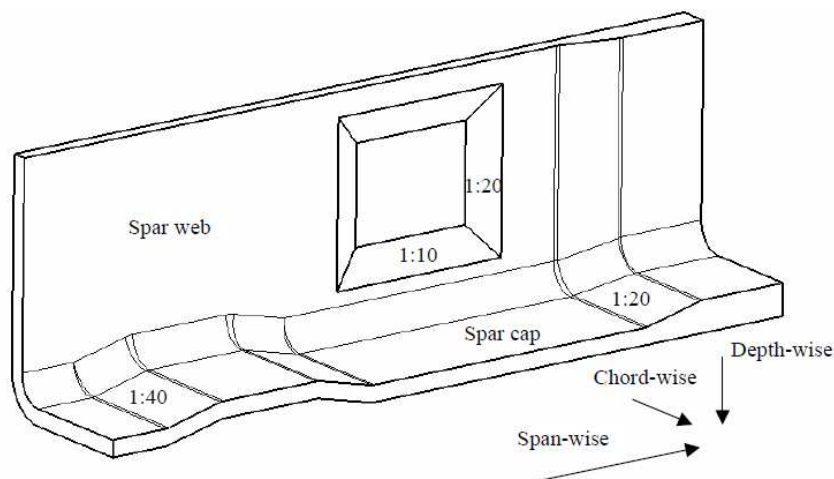


Figure 60 Spar thickness changes

3.2.3.10 Distance from rear spar foot to first stringer foot

The minimum distance from rear spar to first stringer is given by Stress department (see Ref. 2). The distance between rear spar foot and stringer foot should assure a minimum space for riveting the rib when Stress Department considers necessary (if they consider too high the distance between the two elements).

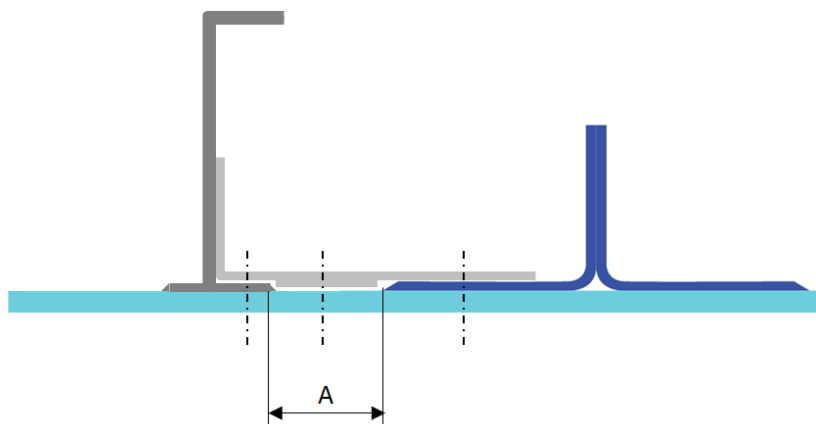


Figure 61 Distance from rear spar foot to first stringer foot scheme

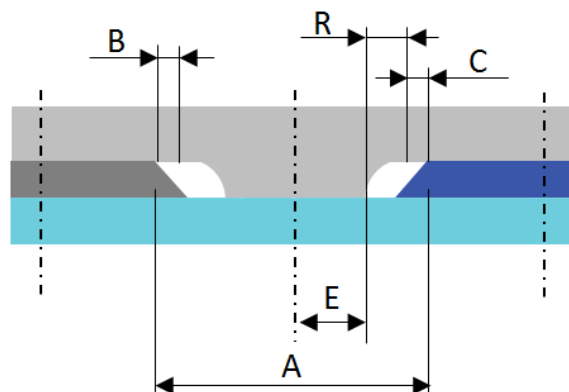


Figure 62 Distance from rear spar to first stringer detail scheme

	MINIMUM DISTANCE FROM REAR SPAR TO FIRST STRINGER				
	B	C	R	E	A
ø fastener	Ttrim+1	Ttrim+3tol+1	4	1,5ø+1	2R+2E+B+C
4,2	2,5	5,5	4	7,3	31
4,8	2,5	5,5	4	8,2	32
6,4	2,5	5,5	4	10,6	37

Figure 63 Minimum distance from rear spar to first stringer

The distances given in the tables do not consider the bevel. When considering it, an additional distance per bevel must be added (additional distance per bevel= thickness/tag32.5°)

3.2.4 Skin

3.2.4.1 Thickness

The maximum skin thickness is limited by manufacturing capabilities: TBD

The minimum skin thickness is limited by lightning strike protection requirements.

3.2.4.2 Drop-offs of ±45° plies

Due to MCL of the AFP machine, for dropping ±45° plies diamond shapes are created.

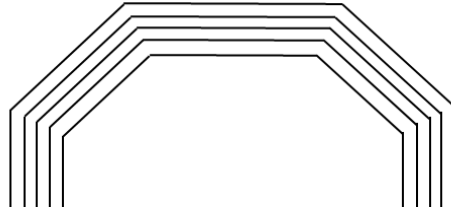


Figure 64 Drop-offs diamond shapes

The size and shape of the contour of these plies are defined by their cutting sequence. For this reason, the sequence chosen should minimize the size of this contour.

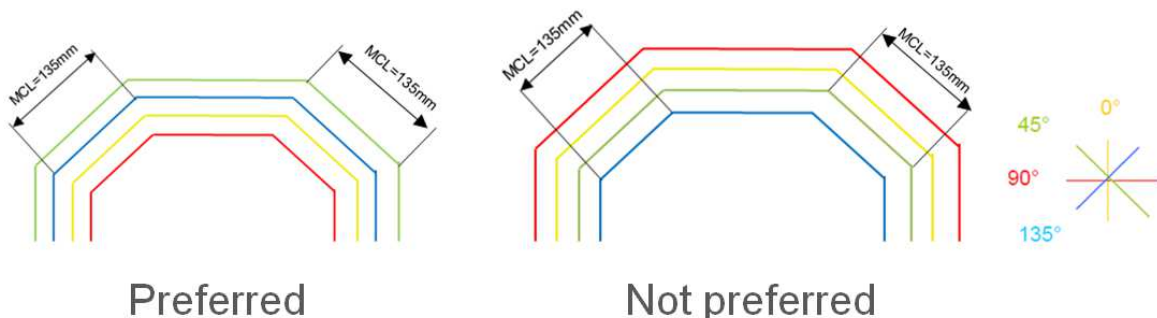


Figure 65 Diamond shape ±45° plies implication

As it is shown in the figure, for the laminate considered, it is recommended to drop the plies of ±45° in the last positions, in order to reduce the overall size of the pad-up.

It must be taken into account that these recommendations can go against general stacking sequence rules. Each specific case should be studied.

3.2.4.3 Slopes under stringers

As a general rule, the slopes under stringers maximum ratio should be 1/20. Skin chordwise thickness changes must not be placed under stringer feet, in order to a correct adaptation of the stringer to the skin.

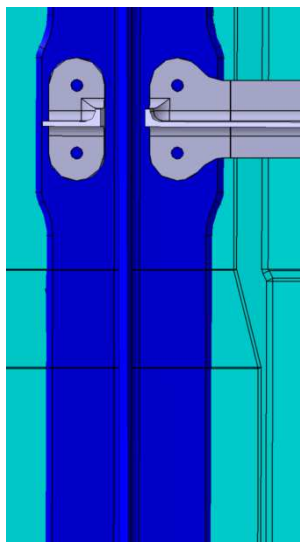


Figure 66 Slopes under stringers

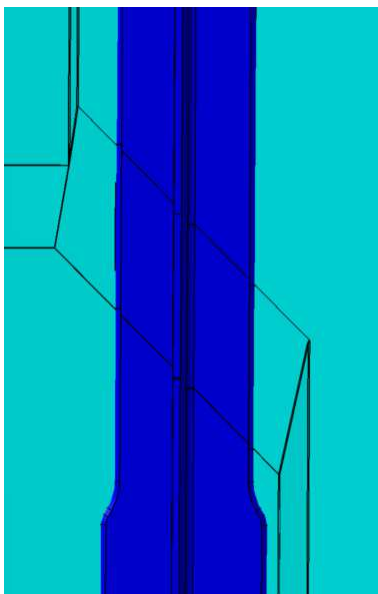


Figure 67 Figure 68 Slopes $\pm 45^\circ$ plies under stringers

It was agreed with Stress Department trying to minimize cutting $\pm 45^\circ$ plies under stringer foot. Only when necessary, a maximum of four $\pm 45^\circ$ plies can be cut. For this situation, the maximum allowable slope ratio is limited to 1/40.

3.2.5 Interfaces

3.2.5.1 Trailing edge interface

For trailing edge interface machined buttstraps riveted to the skin are used. Depending if the trailing edge panel is fixed or removable, the straps have different dimensions. They are fastened to the spar cap and the scheme is the following:

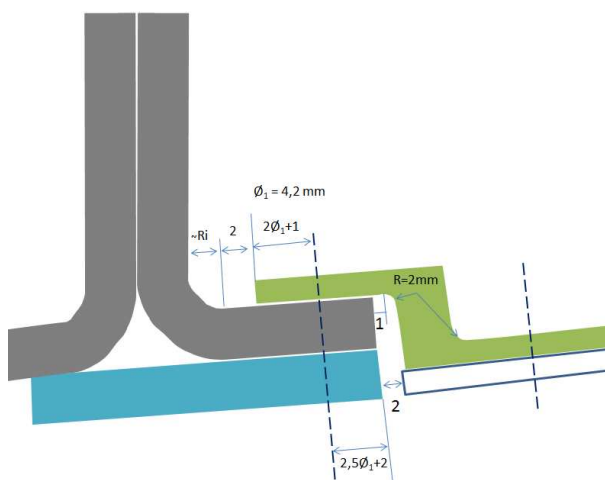


Figure 69 Trailing edge interface

For ribs interface the scheme is the following:

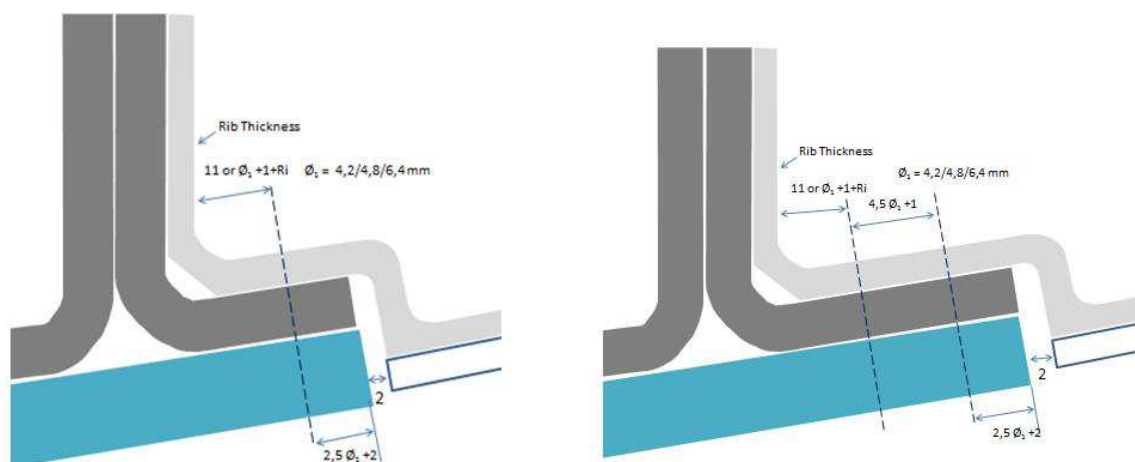


Figure 70 Trailing edge interface rib station scheme for single and double riveted joint.

3.2.5.2 Leading edge interface

For leading edge interface a similar solution of trailing edge has been considered. Machined buttstraps are used. In this case all leading edge panels are removable.

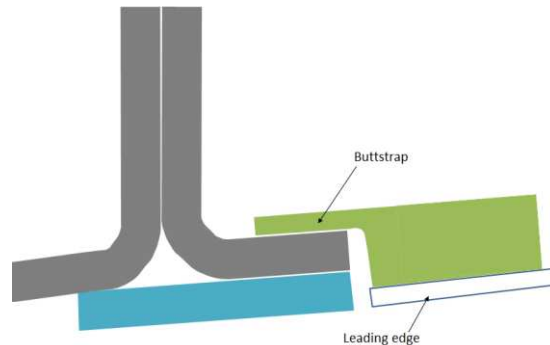


Figure 71 Leading edge interface

3.2.5.3 Accessibility

The minimum size of the cut-out in the spar is limited by accessibility rules considered in Ref. 5.

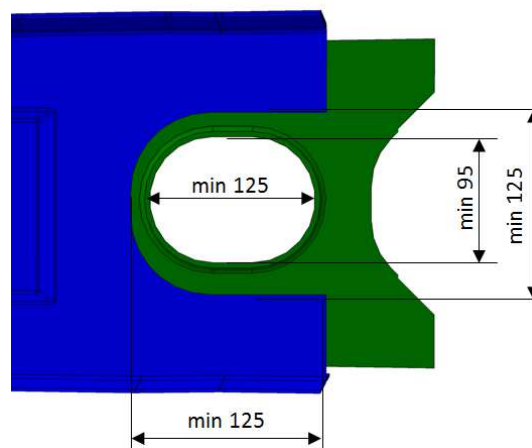


Figure 72 Spar cut-out minimum size

3.2.5.4 Winglet interface

The spars have been cut and a fitting have been added to be riveted to upper and lower skin and to the rib.

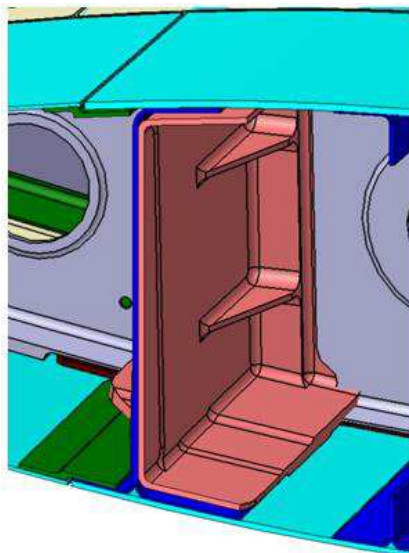


Figure 73 Spar winglet fitting

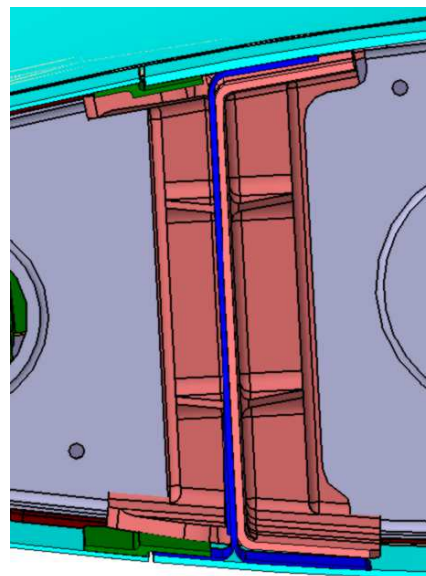


Figure 74 Y8000 rib interface

The fitting has an H shape to join correctly all the parts of the interface.

3.2.5.5 Y0 rib interface

For Y0 rib interface (interface with internal wing box) there are two possible solutions.

- Baseline solution: each stringer has a typical run-out and is joined to the skin by two stringer cleats. The skin has a special element integrated to be joined to the brackets

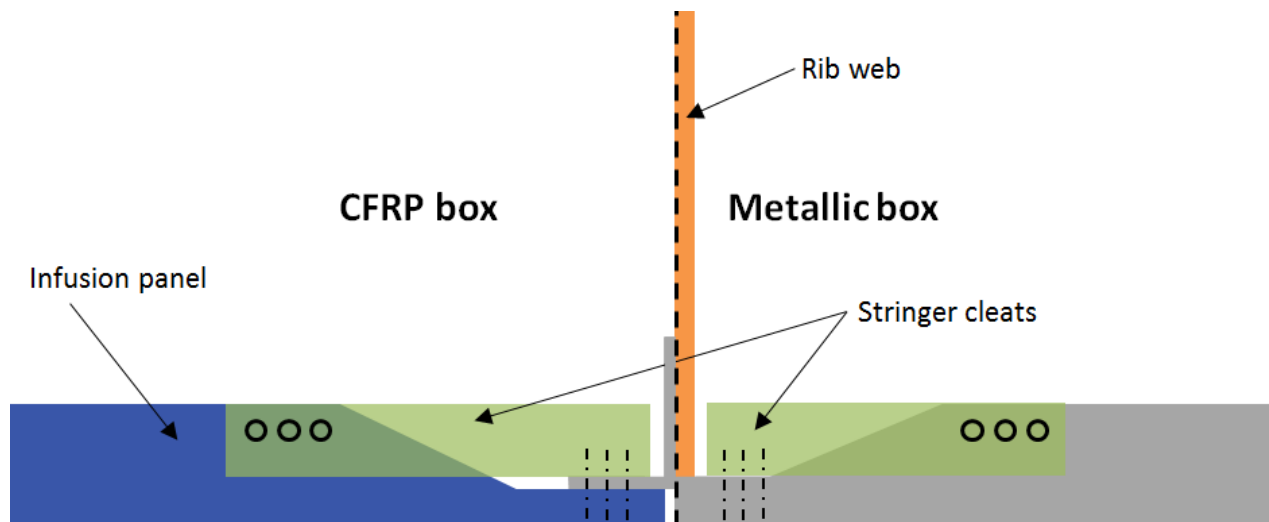


Figure 75 Y0 rib interface baseline solution

-Back-up solution: the stringers run-out and stringer cleats remain the same, but in this case they are joined to a T-shape shear tie

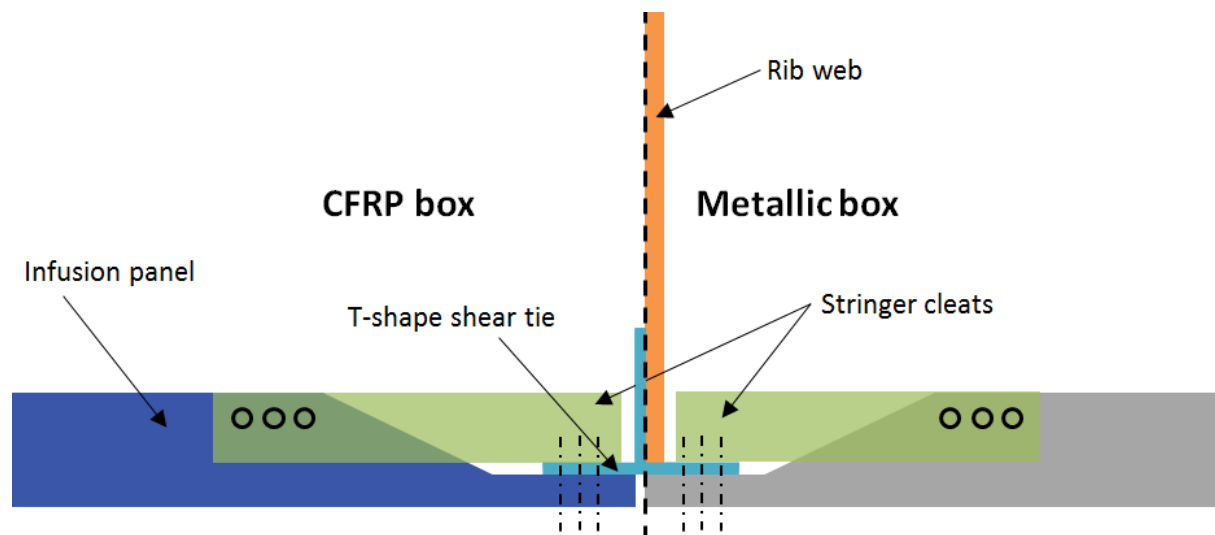


Figure 76 Y0 rib interface back-up solution

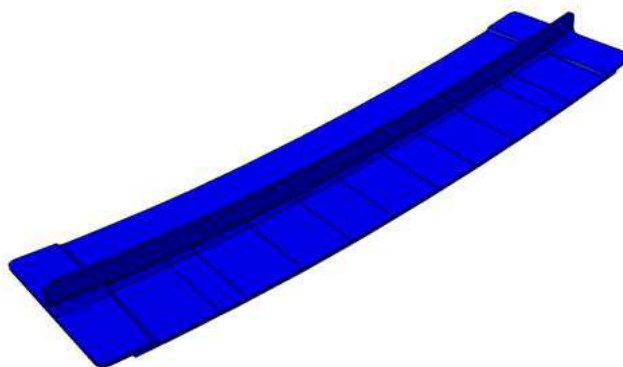


Figure 77 T-shape shear tie

The selected option is yet to be determined.

4. DIMENSIONAL STUDY OF TEST SPECIMENS

4.1 Introduction

The objective of this statistical study is to obtain general tolerances for parts manufactured with AFP and LRI, based on data of existing parts. The data used are from dimensional reports of test specimens of CS2, detail tests and subcomponent tests.

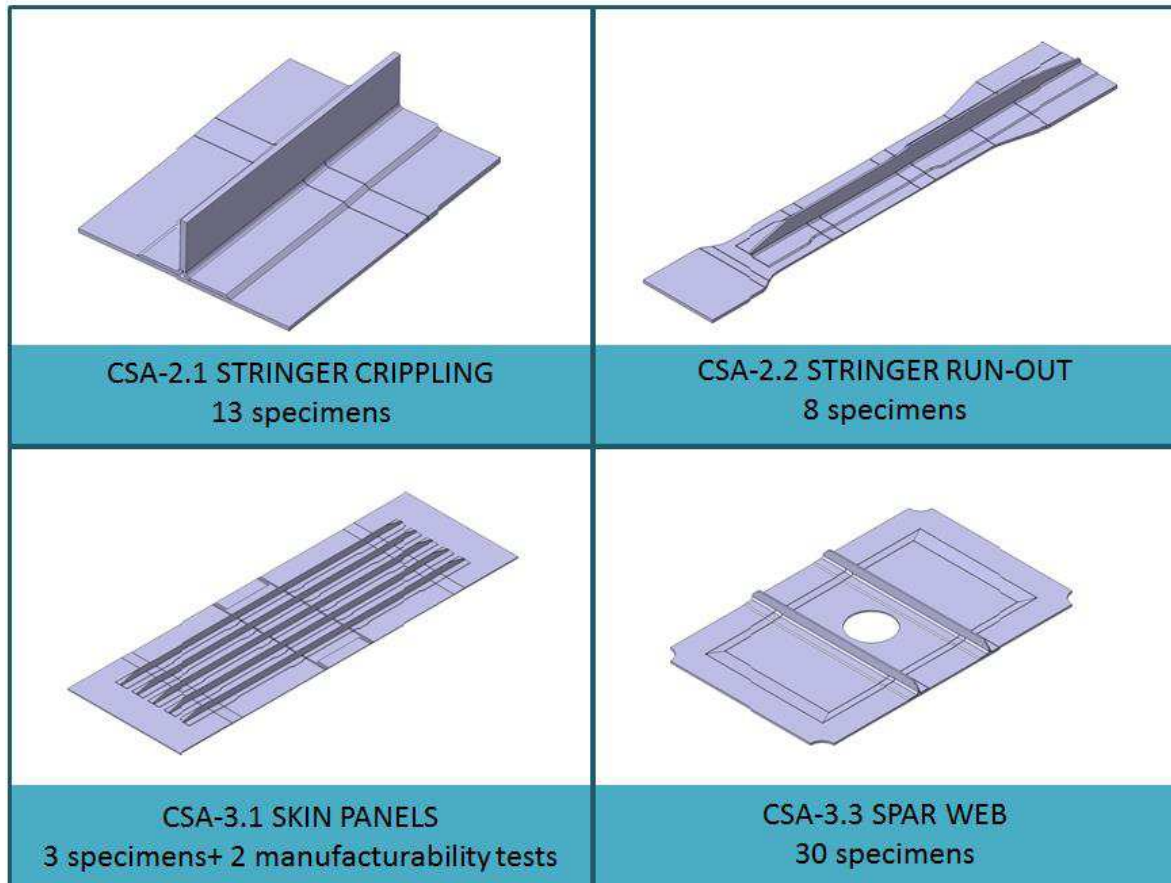


Figure 78 Test specimens CS2

All specimens were manufactured with the same materials:

- Reinforcement: Hexcel Hi-Tape UD210 IMA-12K/12.7mm/V800E
- Thermoset resin: RTM6 (AIMS 05-04-100, IPS 05-04-100-01)

These specimens were specifically designed to study the capabilities of LRI technology and to provide valuable information to specific design details and geometries.

The data from dimensional reports have been ordered by test, property measured and nominal value. For each one, the following data have been compiled and calculated:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of samples	min		max		mean (mm)	% deviation mean vs nom	median (mm)	3 σ			2 σ		
					min value (mm)	% deviation	max value (mm)	% deviation				- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation

Table 15 Standard table used for dimensional studies

-Test information: including the test ID, description of the test, and the property that is being measured.

-Data: including the nominal value and the number of measurements.

-Measured values: including the minimum and maximum values measured, and its corresponding percentage of deviation with respect to the nominal.

-Average values: including the mean, its deviation with respect to the nominal, and the median.

-Deviations: including the values of $\pm 2\sigma$ and $\pm 3\sigma$ and its deviation in respect to the mean. These data are only useful for normal distribution. In a normal distribution, the $(-2\sigma, +2\sigma)$ interval encompasses the 95,46% of the data, and the $(-3\sigma, +3\sigma)$ interval 99,73% of the data.

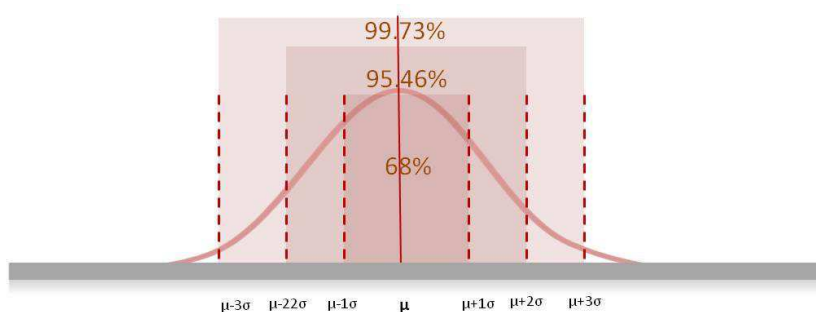


Figure 79 Normal distribution: $\pm 2\sigma$ and $\pm 3\sigma$

The parameters related to normal distribution ($\pm 2\sigma$ and $\pm 3\sigma$) will be only use for statistical studies whose distributions are approximately normally distributed. For that reason, in some cases only absolute values will be studied.

4.2 Thicknesses in infusion tests

4.2.1 Data overview

In infusion specimens the thicknesses obtained are critical features, due to the manufacturing process. One face of the part is highly controlled by the tooling, but the other face finish depends on the bag pressure and other factors.

For thicknesses values there are available data from the four test specimens. There are approximately 2700 data of thicknesses available. Furthermore, thickness has been measured in three differential areas: stringer web, skin and skin+stringer foot. These are the data obtained:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES				DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of sample	min value (mm)	% deviation	max value (mm)	max	mean (mm)	% deviation mean vs noi	median (mm)	- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation	2 σ
CS A-2.1	Stringer Crippling	str web	3.53	42	3.54	-0.28%	4.04	14.45%	3.71	5.10%	3.67	3.31	4.11	10.78%	3.45	3.98	7.28%	
CS A-2.1	Stringer Crippling	str web	4.31	36	4.32	-0.23%	4.64	7.66%	4.49	4.18%	4.48	4.17	4.82	7.35%	4.27	4.71	4.90%	
CS A-2.1	Stringer Crippling	skin	2.35	84	2.32	1.28%	2.74	16.60%	2.50	6.38%	2.52	2.19	2.82	12.80%	2.29	2.71	7.54%	
CS A-2.1	Stringer Crippling	skin	2.74	98	2.68	2.19%	3.16	15.33%	2.93	6.93%	2.94	2.64	3.21	9.56%	2.74	3.12	6.12%	
CS A-2.1	Stringer Crippling	skin	3.72	28	3.84	-3.23%	4.18	12.37%	3.97	6.72%	3.97	3.73	4.22	6.30%	3.81	4.13	4.03%	
CS A-2.1	Stringer Crippling	skin + str foot	4.12	28	4.02	2.31%	4.36	5.95%	4.19	1.82%	4.21	3.88	4.51	7.64%	3.98	4.40	5.01%	
CS A-2.1	Stringer Crippling	skin + str foot	4.51	14	4.44	1.44%	4.74	5.22%	4.60	2.11%	4.60	4.29	4.91	6.74%	4.39	4.81	4.57%	
CS A-2.1	Stringer Crippling	skin + str foot	4.90	24	4.94	-0.92%	5.48	11.95%	5.15	5.21%	5.10	4.71	5.58	8.35%	4.85	5.44	5.63%	
CS A-2.1	Stringer Crippling	skin + str foot	5.88	12	5.8	1.28%	6.12	4.17%	5.98	1.79%	5.99	5.72	6.25	4.52%	5.81	6.16	3.01%	
CS A-2.2	Stringer Run-out	skin	2.35	48	2.50	-6.38%	2.80	19.15%	2.64	12.34%	2.62	2.41	2.87	8.71%	2.49	2.79	6.49%	
CS A-2.2	Stringer Run-out	skin	3.72	48	4.00	-7.53%	4.44	19.35%	4.17	12.10%	4.17	3.85	4.5	7.91%	3.96	4.39	5.28%	
CS A-2.2	Stringer Run-out	skin	3.92	48	4.04	-3.06%	4.52	15.31%	4.31	9.95%	4.31	3.94	4.68	8.58%	4.06	4.56	5.80%	
CS A-2.2	Stringer Run-out	skin	5.49	36	5.70	-3.83%	6.02	9.65%	5.86	6.74%	5.85	5.65	6.08	3.75%	5.72	6.01	2.74%	
CS A-2.2	Stringer Run-out	skin	6.27	64	6.42	-2.39%	7.62	21.53%	6.71	7.02%	6.68	6.07	7.36	9.69%	6.28	7.14	6.89%	
CS A-2.2	Stringer Run-out	skin + str foot	4.12	32	4.10	0.49%	4.72	14.56%	4.46	8.25%	4.51	3.97	4.96	11.21%	4.13	4.79	7.40%	
CS A-2.2	Stringer Run-out	skin + str foot	5.49	24	5.64	-2.73%	5.98	8.93%	5.83	6.19%	5.84	5.57	6.09	4.46%	5.66	6.00	2.92%	
CS A-2.2	Stringer Run-out	skin + str foot	5.69	40	5.52	2.95%	6.26	10.02%	5.96	4.75%	6.03	5.25	6.67	11.91%	5.49	6.43	7.89%	
CS A-2.2	Stringer Run-out	skin + str foot	8.04	32	7.89	1.87%	8.46	5.22%	8.25	2.61%	8.28	7.87	8.64	4.73%	8.00	8.51	3.15%	
CS A-2.2	Stringer Run-out	str web	3.53	62	3.64	-3.12%	4.18	18.41%	3.85	9.07%	3.82	3.50	4.20	9.09%	3.62	4.08	5.97%	
CS A-3.1	Skin Panel	str web	3.53	150	3.41	3.28%	4.58	29.76%	3.94	11.68%	3.91	3.24	4.64	17.77%	3.48	4.40	11.68%	
CS A-3.1	Skin Panel	skin	2.35	18	2.63	-11.82%	2.86	21.60%	2.72	15.65%	2.70	2.54	2.90	6.62%	2.60	2.84	4.41%	
CS A-3.1	Skin Panel	skin	2.74	54	2.89	-5.32%	3.45	25.73%	3.11	13.34%	3.10	2.83	3.38	8.68%	2.93	3.29	5.79%	
CS A-3.1	Skin Panel	skin	6.27	60	6.41	-2.20%	6.94	10.65%	6.7	6.82%	6.67	6.28	7.12	6.27%	6.42	6.98	4.18%	
CS A-3.1	Skin Panel	skin + str foot	4.12	30	4.48	-8.75%	5.25	27.45%	4.80	16.62%	4.81	4.12	5.48	14.17%	4.35	5.26	9.58%	
CS A-3.1	Skin Panel	skin + str foot	4.51	60	4.74	-5.15%	5.30	17.57%	4.95	9.80%	4.91	4.52	5.39	8.89%	4.66	5.24	5.86%	
CS A-3.1	Skin Panel	skin + str foot	8.04	60	8.25	-2.66%	9.02	12.24%	8.58	6.77%	8.54	8.06	9.09	5.94%	8.24	8.92	3.96%	
CS A-3.3	Spar Web	str web	2.74	174	2.68	2.19%	3.62	32.12%	3.13	14.23%	3.12	2.63	3.63	15.97%	2.80	3.47	10.86%	
CS A-3.3	Spar Web	str web	3.53	186	3.34	5.38%	4.32	22.38%	3.85	9.07%	3.86	3.31	4.29	13.77%	3.49	4.20	9.09%	
CS A-3.3	Spar Web	skin	2.35	178	2.32	1.28%	3.1	31.91%	2.59	10.21%	2.56	2.19	2.99	15.44%	2.32	2.86	11.72%	
CS A-3.3	Spar Web	skin	3.53	180	3.44	2.55%	4.34	22.95%	3.76	6.52%	3.72	3.21	4.32	14.89%	3.40	4.14	11.29%	
CS A-3.3	Spar Web	skin	3.72	190	3.72	0.00%	4.36	17.20%	3.98	6.99%	3.98	3.62	4.34	9.05%	3.74	4.22	6.03%	
CS A-3.3	Spar Web	skin	5.49	190	5.32	3.10%	6.26	14.03%	5.75	4.74%	5.74	5.2	6.31	9.74%	5.38	6.13	6.79%	
CS A-3.3	Spar Web	skin + str foot	3.72	176	3.84	-3.23%	4.22	13.44%	3.99	7.26%	3.98	3.74	4.25	6.52%	3.82	4.16	4.26%	
CS A-3.3	Spar Web	skin + str foot	5.30	180	5.3	-0.09%	5.78	9.16%	5.52	4.25%	5.52	5.19	5.86	6.16%	5.30	5.74	3.99%	

Table 16 Thicknesses in infusion test data

In this case the value used for considering the deviation acceptable or not is $\pm 5\%$. This is the typical value of general thicknesses tolerances for prepreg technology. For the moment it has been used as a reference.

There is a considerable difference between the size of stringer crippling (CS A-2.1) and the rest of the tests.

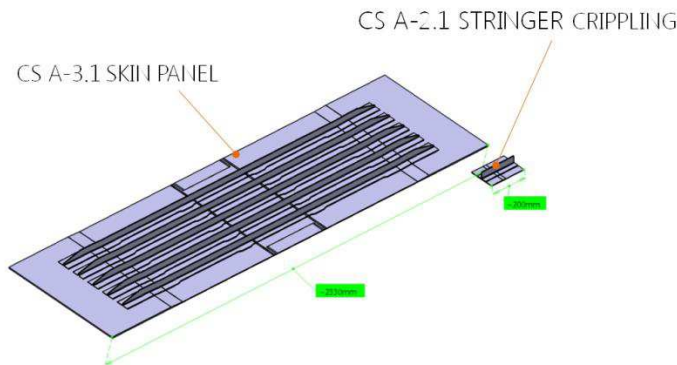


Figure 80 Size difference between stringer crippling and skin panel

The size of the specimen can affect to thicknesses tolerances, as can be seen in the next table, where the percentage of deviation of the mean with respect to the nominal value of each type of test specimen is shown:

	% deviation mean vs nom
Stringer Crippling	4,5%
Spar Web	7,9%
Stringer Run-out	7,9%
Skin Panel	11,6%

+ size of specimen
+ deviation

Table 17 Percentage of deviation of the mean in respect to the nominal value in thicknesses studies

The deviation of the mean versus the nominal of the stringer crippling is much lower than in the others. For this reason, the stringer crippling is not considered representative of the manufacturing process and the data obtained are not being considered for this analysis.

4.2.2 Stringer web thickness

The data from stringer web thickness ordered by test are:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of samples	min value (mm)	% deviation	max value (mm)	% deviation	mean (mm)	% deviation mean vs nom	median (mm)	3 σ			2 σ		
												- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
CS A-2.2	Stringer Run-out	str web	3,53	62	3,64	-3,12%	4,18	18,41%	3,85	9,07%	3,82	3,50	4,20	9,09%	3,62	4,08	5,97%
CS A-3.1	Skin Panel	str web	3,53	150	3,41	3,29%	4,58	29,76%	3,94	11,68%	3,91	3,24	4,64	17,77%	3,48	4,40	11,68%
CS A-3.3	Spar Web	str web	2,74	174	2,68	2,19%	3,62	32,12%	3,13	14,23%	3,12	2,63	3,63	15,97%	2,80	3,47	10,86%
CS A-3.3	Spar Web	str web	3,53	186	3,34	5,38%	4,32	22,38%	3,85	9,07%	3,86	3,31	4,38	13,77%	3,49	4,20	9,09%

Table 18 Stringer web thickness data

The % of deviation of the mean versus the nominal for 2,74 mm thickness is 14,23%, and for 3,53 mm thickness is 9,94% (similar deviation obtained from 3 different tests). The average deviation is 11,01%.

In the next figure it can be observed that the data distribution is approximately normally distributed:

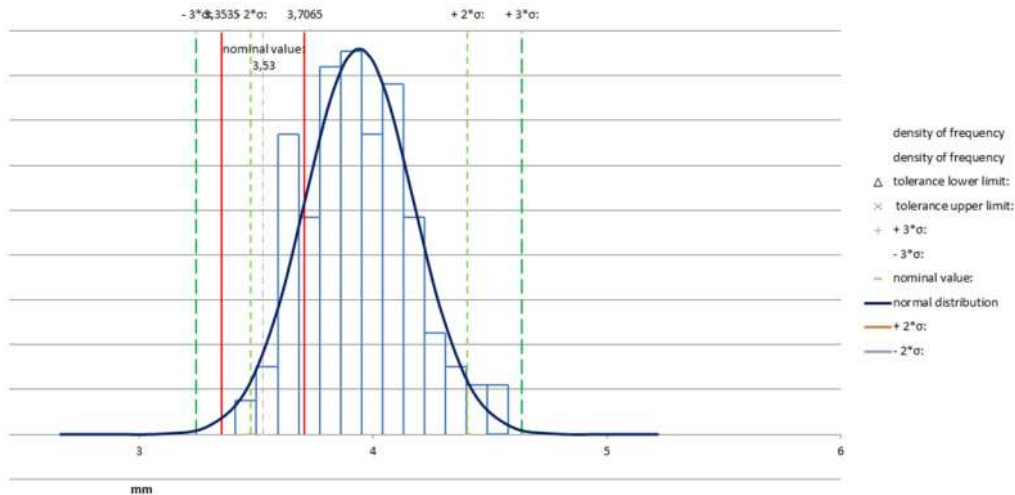


Figure 81 Normal distribution for stringer web thicknesses (nominal 3,53 mm)

4.2.3 Skin thickness

The data from skin thickness ordered by test are:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of sample	min value (mm)	% deviation	max value (mm)	% deviation	mean (mm)	% deviation mean vs nom	median (mm)	3 σ			2 σ		
												- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
CS A-2.2	Stringer Run-out	skin	2,35	48	2,50	-6,38%	2,80	19,15%	2,64	12,34%	2,62	2,41	2,87	8,71%	2,49	2,79	6,49%
CS A-2.2	Stringer Run-out	skin	3,72	48	4,00	-7,53%	4,44	19,35%	4,17	12,10%	4,17	3,85	4,5	7,91%	3,96	4,39	5,28%
CS A-2.2	Stringer Run-out	skin	3,92	48	4,04	-3,06%	4,52	15,31%	4,31	9,95%	4,31	3,94	4,68	8,58%	4,06	4,56	5,80%
CS A-2.2	Stringer Run-out	skin	5,49	36	5,70	-3,83%	6,02	9,65%	5,86	6,74%	5,85	5,65	6,08	3,75%	5,72	6,01	2,74%
CS A-2.2	Stringer Run-out	skin	6,27	64	6,42	-2,39%	7,62	21,53%	6,71	7,02%	6,68	6,07	7,36	9,69%	6,28	7,14	6,89%
CS A-3.1	Skin Panel	skin	2,35	18	2,63	-11,82%	2,86	21,60%	2,72	15,65%	2,70	2,54	2,90	6,62%	2,60	2,84	4,41%
CS A-3.1	Skin Panel	skin	2,74	54	2,89	-5,32%	3,45	25,73%	3,11	13,34%	3,10	2,83	3,38	8,68%	2,93	3,29	5,79%
CS A-3.1	Skin Panel	skin	6,27	60	6,41	-2,20%	6,94	10,65%	6,7	6,82%	6,67	6,28	7,12	6,27%	6,42	6,98	4,18%
CS A-3.3	Spar Web	skin	2,35	178	2,32	1,28%	3,1	31,91%	2,59	10,21%	2,56	2,19	2,99	15,44%	2,32	2,86	11,72%
CS A-3.3	Spar Web	skin	3,53	180	3,44	2,55%	4,34	22,95%	3,76	6,52%	3,72	3,21	4,32	14,89%	3,40	4,14	11,29%
CS A-3.3	Spar Web	skin	3,72	190	3,72	0,00%	4,36	17,20%	3,98	6,99%	3,98	3,62	4,34	9,05%	3,74	4,22	6,03%
CS A-3.3	Spar Web	skin	5,49	190	5,32	3,10%	6,26	14,03%	5,75	4,74%	5,74	5,2	6,31	9,74%	5,38	6,13	6,79%

Table 19 Skin thickness data (ordered by test)

The data have been also ordered by thickness value, to see if there is a relation between the value of the thickness and the deviations. It will also be useful to detect out of average data, and some possible problems in the measurements or in some concrete specimens.

The data from skin thickness ordered by nominal value are:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of sample	min value (mm)	% deviation	max value (mm)	% deviation	mean (mm)	% deviation mean vs nom	median (mm)	3 σ			2 σ		
												- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
CS A-2.2	Stringer Run-out	skin	2,35	48	2,50	-6,38%	2,80	19,15%	2,64	12,34%	2,62	2,41	2,87	8,71%	2,49	2,79	6,49%
CS A-3.3	Spar Web	skin	2,35	178	2,32	1,28%	3,1	31,91%	2,59	10,21%	2,56	2,19	2,99	15,44%	2,32	2,86	11,72%
CS A-3.1	Skin Panel	skin	2,35	18	2,63	-11,82%	2,86	21,60%	2,72	15,65%	2,70	2,54	2,90	6,62%	2,60	2,84	4,41%
CS A-3.1	Skin Panel	skin	2,74	54	2,89	-5,32%	3,45	25,73%	3,11	13,34%	3,10	2,83	3,38	8,68%	2,93	3,29	5,79%
CS A-3.3	Spar Web	skin	3,53	180	3,44	2,55%	4,34	22,95%	3,76	6,52%	3,72	3,21	4,32	14,89%	3,40	4,14	11,29%
CS A-2.2	Stringer Run-out	skin	3,72	48	4,00	-7,53%	4,44	19,35%	4,17	12,10%	4,17	3,85	4,5	7,91%	3,96	4,39	5,28%
CS A-3.3	Spar Web	skin	3,72	190	3,72	0,00%	4,36	17,20%	3,98	6,99%	3,98	3,62	4,34	9,05%	3,74	4,22	6,03%
CS A-2.2	Stringer Run-out	skin	3,92	48	4,04	-3,06%	4,52	15,31%	4,31	9,95%	4,31	3,94	4,68	8,58%	4,06	4,56	5,80%
CS A-2.2	Stringer Run-out	skin	5,49	36	5,70	-3,83%	6,02	9,65%	5,86	6,74%	5,85	5,65	6,08	3,75%	5,72	6,01	2,74%
CS A-3.3	Spar Web	skin	5,49	190	5,32	3,10%	6,26	14,03%	5,75	4,74%	5,74	5,2	6,31	9,74%	5,38	6,13	6,79%
CS A-2.2	Stringer Run-out	skin	6,27	64	6,42	-2,39%	7,62	21,53%	6,71	7,02%	6,68	6,07	7,36	9,69%	6,28	7,14	6,89%
CS A-3.1	Skin Panel	skin	6,27	60	6,41	-2,20%	6,94	10,65%	6,7	6,82%	6,67	6,28	7,12	6,27%	6,42	6,98	4,18%

Table 20 Skin thickness data (ordered by nominal value)

As in the previous case, the data distribution is approximately normally distributed.

In the next figure it can be observed that 4 thicknesses deviations are significantly lower than the others. These 4 nominal thicknesses correspond to the spar web tests (CS A-3.3).

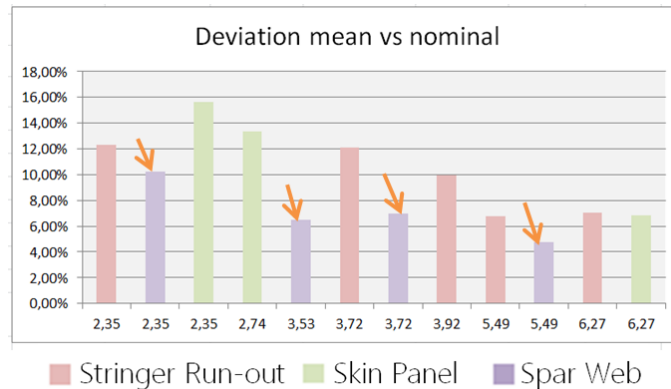


Figure 82 Deviations mean vs nominal for skin thickness

It seems that the differences may be due to the size difference or a different manufacturing process. Therefore, measures of skin thickness from Spar Web Test are not being considered.

Without the Spar Web Test contributions, the graph looks like:

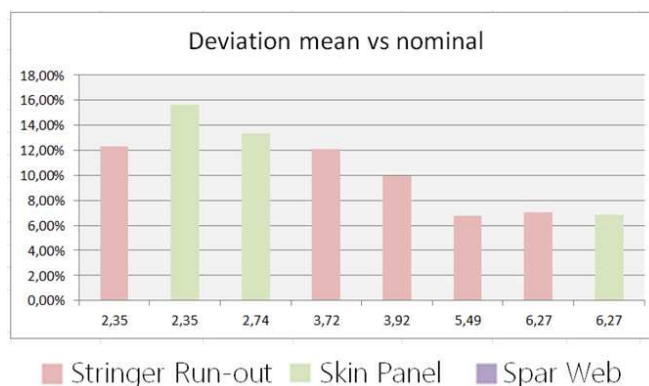


Figure 83 Deviations mean vs nominal for skin thickness (without Spar Web Test contributions)

There are still two deviations higher than the others. They correspond to skin panels tests. The size of these specimens is bigger, so it is more difficult to control the thickness. The difference is reasonable, so they are being considered for the study.

The average deviation is 10,49%, very similar value to the mean of stringer web thickness (11,01 %).

4.2.4 Skin + stringer foot thickness

The data from skin + stringer foot thickness ordered by nominal value are:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
test ID	description	property measured	nominal value (mm)	number of sample	min value (mm)	% deviation	max value (mm)	% deviation	mean (mm)	% deviation mean vs noi	median (mm)	3 σ			2 σ		
												- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
CS A-3.3	Spar Web	skin + str foot	3,72	176	3,84	-3,23%	4,22	13,44%	3,99	7,26%	3,98	3,74	4,25	6,52%	3,82	4,16	4,26%
CS A-3.1	Skin Panel	skin + str foot	4,12	30	4,48	-8,75%	5,25	27,45%	4,80	16,62%	4,81	4,12	5,48	14,17%	4,35	5,26	9,58%
CS A-2.2	Stringer Run-out	skin + str foot	4,12	32	4,10	0,49%	4,72	14,56%	4,46	8,25%	4,51	3,97	4,96	11,21%	4,13	4,79	7,40%
CS A-3.1	Skin Panel	skin + str foot	4,51	60	4,74	-5,15%	5,30	17,57%	4,95	9,80%	4,91	4,52	5,39	8,89%	4,66	5,24	5,86%
CS A-3.3	Spar Web	skin + str foot	5,30	180	5,3	-0,09%	5,78	9,16%	5,52	4,25%	5,52	5,19	5,86	6,16%	5,30	5,74	3,99%
CS A-2.2	Stringer Run-out	skin + str foot	5,49	24	5,64	-2,73%	5,98	8,93%	5,83	6,19%	5,84	5,57	6,09	4,46%	5,66	6,00	2,92%
CS A-2.2	Stringer Run-out	skin + str foot	5,69	40	5,52	2,99%	6,26	10,02%	5,96	4,75%	6,03	5,25	6,67	11,91%	5,49	6,43	7,89%
CS A-3.1	Skin Panel	skin + str foot	8,04	60	8,25	-2,66%	9,02	12,24%	8,58	6,77%	8,54	8,06	9,09	5,94%	8,24	8,92	3,96%
CS A-2.2	Stringer Run-out	skin + str foot	8,04	32	7,89	1,87%	8,46	5,22%	8,25	2,61%	8,28	7,87	8,64	4,73%	8,00	8,51	3,15%

Figure 84 Skin+stringer foot thickness data

As in the previous case, the data distribution is approximately normally distributed.

It can be observed that one thickness deviation is significantly higher than the others. It is approximately two times more than the same thickness measured in stringer run-out test).

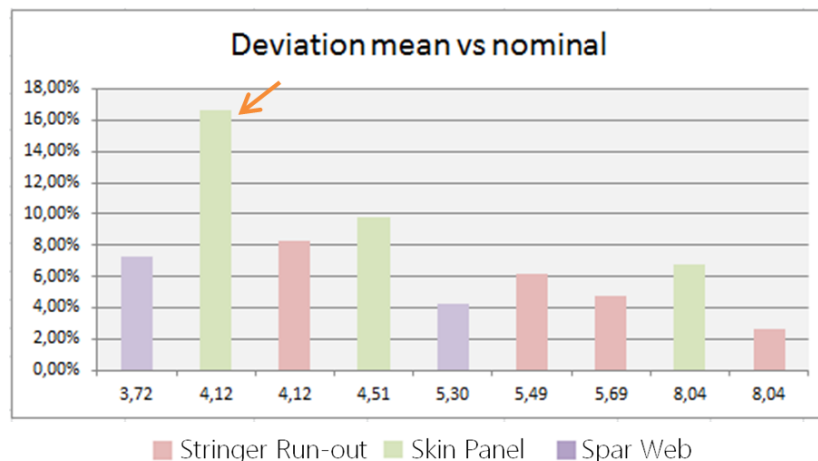


Figure 85 Deviations mean vs nominal for skin and stringer foot thickness

This nominal thickness corresponds to the skin panel tests (CS A-3.1), specifically to data of skin+stringer foot thickness, in the central area (minimum thickness). This could be due to a problem in the manufacturing process, which should be addressed by the manufacturing team. Therefore the nominal of 4,12 mm from skin panel tests is not being considered in this study.

Other two thicknesses deviations corresponding to the skin panel tests (CS A-3.1) are slightly higher than the others. These data of 4,51 mm and 8,04 mm are being considered because the increase is reasonable, taking into account that the skin panels are bigger than the other tests, so the thickness is more difficult to control.

If the thickness of 4,12 mm is not considered, the graph looks like:

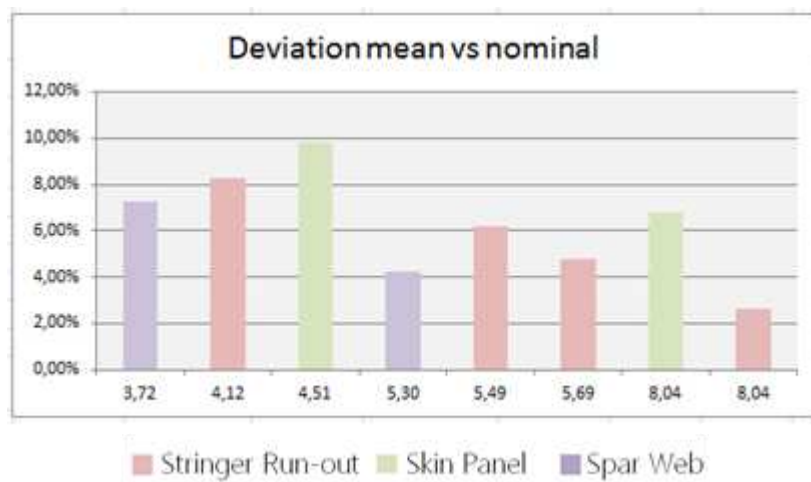


Figure 86 Deviations mean vs nominal for skin and stringer foot thickness (without 4,12 nominal)

The mean of deviation in this case is 6,24%. This value is considerably lower than the stringer web thickness (11,01%) and skin thickness (10,49%) mean of deviation.

4.2.5 Comparative study of all data

For each thickness considered, some common characteristics have been observed:

- Minimum value: quite near or even higher than the nominal value
- Maximum value: over 5% deviation (in many cases well above)
- Mean: in all cases over the nominal value
- Median: similar to mean (symmetric distribution)
- Data approximately normally distributed
- Deviations 2σ and 3σ : deviation with respect to the mean over 5% in most cases

In summary, all areas/components present over-thickness and the tolerance ranges are wide.

In the chart the deviations of the mean versus the nominal value of data from all types of specimens are shown, ordered by nominal value (taking out the nominal values previously considered not representative):

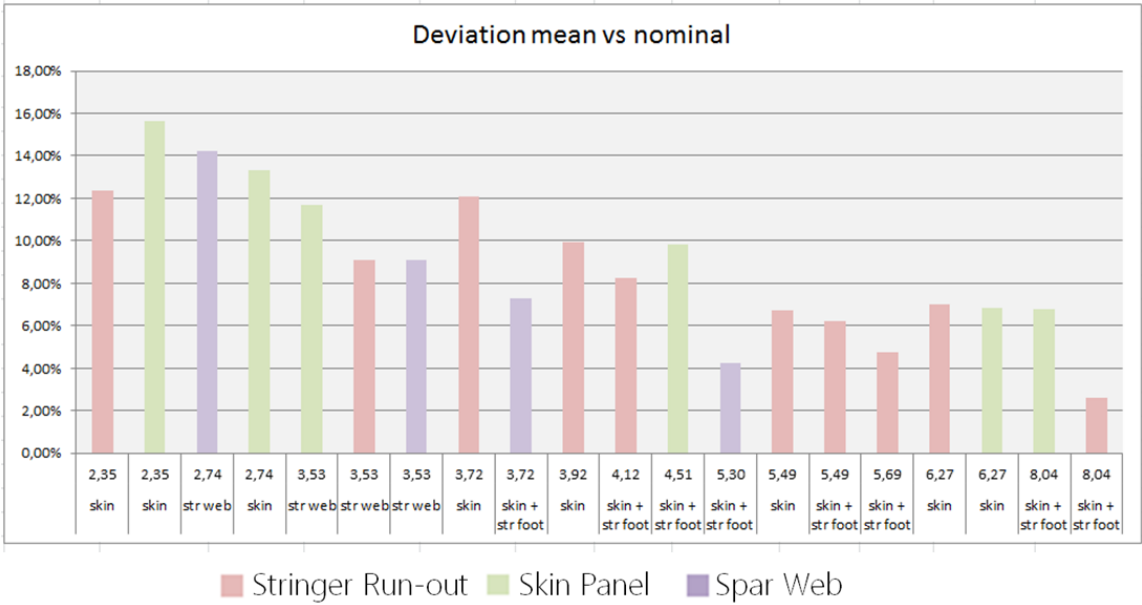


Figure 87 Deviations mean vs nominal of all properties and tests (ordered by nominal value)

The overthickness is a fact in all cases, but the value of it varies. There is a slight relation between the value of the thickness and the deviation. For thicker thicknesses the value of overthickness decreases. However, it doesn't follow a clear scheme to determine a concrete overthickness.

When comparing the overthickness per property measured, the following data are obtained:

Property measured	% deviation mean vs nom
Stringer Web	11,01%
Skin	10,49%
Skin + Stringer foot	6,24%
	MEAN= 8,89 %*

*Mean calculated from all deviations from each nominal, not from each property measured

Table 21 Deviation of mean vs nominal per property measured

It can be observed that two of the properties measured have very similar deviations (approximately 11%).

However, the skin + stringer foot measurements have lower deviations. It could be due to the manufacturing process. There are some zones in which caul plates are used. The effect of the caul plates turns into more controlled thicknesses and therefore less deviation.

Although caul plates are also used for stringer webs, the effect in this area is not the same, as the pressure applied by the bag may be lower than on stringer feet.

4.2.6 Study of new nominal

Tolerances with respect to current nominal values are too big to be considered for design. A possible solution to reduce the tolerance range is to increase the nominal value of cured ply thickness.

Actual thickness per cured ply= 0,196 mm

It has been evaluated the effect of a possible higher nominal value (+5%, +6%, +7%, +8%, +9%, +10%).

Test ID	Test	Measured feature	+5%			+6%			+7%			+8%			+9%			+10%		
			new nominal	% deviation to -2σ	% deviation to +2σ	new nominal	% deviation to -2σ	% deviation to +2σ	new nominal	% deviation to -2σ	% deviation to +2σ	new nominal	% deviation to -2σ	% deviation to +2σ	new nominal	% deviation to -2σ	% deviation to +2σ	new nominal	% deviation to -2σ	% deviation to +2σ
CS A-2.2	Stringer Run-out	skin	2,47	0,9%	13,07%	2,49	0,0%	12,00%	2,51	1,0%	10,96%	2,54	1,9%	9,93%	2,56	2,8%	8,92%	2,59	3,7%	7,93%
CS A-3.1	Skin Panel	skin	2,47	5,3%	13,00%	2,49	4,3%	13,91%	2,52	3,3%	12,85%	2,54	2,4%	11,80%	2,56	1,4%	10,78%	2,59	0,3%	9,77%
CS A-3.3	Spar Web	str web	2,88	2,7%	20,61%	2,90	3,6%	19,47%	2,93	4,5%	18,36%	2,96	5,4%	17,26%	2,99	6,2%	16,19%	3,01	7,1%	15,13%
CS A-3.1	Skin Panel	skin	2,88	1,7%	14,19%	2,91	0,7%	13,11%	2,94	0,2%	12,05%	2,96	1,1%	11,02%	2,99	2,0%	10,00%	3,02	2,9%	9,00%
CS A-3.1	Skin Panel	str web	3,70	6,1%	18,78%	3,74	6,9%	17,66%	3,77	7,8%	16,56%	3,81	8,7%	15,48%	3,85	9,5%	14,42%	3,88	10,3%	13,38%
CS A-2.2	Stringer Run-out	str web	3,71	2,3%	10,08%	3,74	3,3%	9,04%	3,78	4,2%	8,02%	3,81	5,0%	7,02%	3,85	5,9%	6,04%	3,88	6,8%	5,07%
CS A-3.3	Spar Web	str web	3,71	5,8%	13,31%	3,74	6,7%	12,25%	3,78	7,6%	11,20%	3,81	8,5%	10,17%	3,85	9,3%	9,16%	3,88	10,1%	8,16%
CS A-2.2	Stringer Run-out	skin	3,91	1,4%	12,39%	3,94	0,4%	11,33%	3,98	0,5%	10,29%	4,02	1,4%	9,27%	4,05	2,3%	8,27%	4,09	3,2%	7,28%
CS A-3.3	Spar Web	skin + str foot	3,91	2,2%	6,50%	3,94	3,1%	5,50%	3,98	4,0%	4,51%	4,02	4,9%	3,54%	4,05	5,8%	2,59%	4,09	6,6%	1,66%
CS A-2.2	Stringer Run-out	skin	4,12	1,4%	10,79%	4,16	2,3%	9,74%	4,19	3,2%	8,72%	4,23	4,1%	7,71%	4,27	5,0%	6,72%	4,31	5,8%	5,75%
CS A-2.2	Stringer Run-out	skin + str foot	4,33	4,5%	10,73%	4,37	5,4%	9,68%	4,41	6,3%	8,66%	4,45	7,2%	7,65%	4,49	8,0%	6,66%	4,53	8,9%	5,69%
CS A-3.1	Skin Panel	skin + str foot	4,73	1,6%	10,70%	4,78	2,5%	9,66%	4,82	3,4%	8,63%	4,87	4,3%	7,63%	4,91	5,2%	6,64%	4,96	6,0%	5,67%
CS A-3.3	Spar Web	skin + str foot	5,56	4,7%	3,24%	5,61	5,6%	2,27%	5,67	6,5%	1,31%	5,72	7,3%	0,37%	5,77	8,2%	0,55%	5,82	9,0%	1,45%
CS A-2.2	Stringer Run-out	skin	5,76	0,8%	4,26%	5,82	1,7%	3,28%	5,87	2,6%	2,31%	5,93	3,5%	1,36%	5,98	4,4%	0,43%	6,04	5,3%	0,48%
CS A-2.2	Stringer Run-out	skin + str foot	5,76	1,8%	4,09%	5,82	2,7%	3,10%	5,87	3,6%	2,14%	5,93	4,5%	1,19%	5,98	5,4%	0,27%	6,04	6,3%	0,65%
CS A-2.2	Stringer Run-out	skin + str foot	5,97	8,1%	7,62%	6,03	9,0%	6,61%	6,09	9,8%	5,61%	6,15	10,7%	4,63%	6,20	11,5%	3,67%	6,26	12,3%	2,73%
CS A-2.2	Stringer Run-out	skin	6,58	4,6%	8,45%	6,65	5,5%	7,43%	6,71	6,4%	6,43%	6,77	7,3%	5,44%	6,83	8,1%	4,47%	6,90	8,9%	3,52%
CS A-3.1	Skin Panel	skin	6,59	2,5%	5,99%	6,65	3,4%	4,99%	6,71	4,3%	4,01%	6,77	5,2%	3,04%	6,84	6,1%	2,10%	6,90	6,9%	1,17%
CS A-3.1	Skin Panel	skin + str foot	8,44	2,3%	5,71%	8,52	3,3%	4,72%	8,60	4,2%	3,74%	8,68	5,1%	2,78%	8,76	5,9%	1,84%	8,84	6,8%	0,91%
CS A-2.2	Stringer Run-out	skin + str foot	8,44	5,2%	0,81%	8,52	6,1%	0,15%	8,60	7,0%	1,08%	8,68	7,9%	1,99%	8,76	8,7%	2,89%	8,84	9,5%	3,78%

Table 22 Study of effect of a new nominal value

However, as it can be seen in the previous table, with any of this new nominal the resulting tolerance ranges are still above 5% in many cases. None of these higher nominal values provide a considerably better result. For this reason it has been agreed to maintain the original nominal value.

4.2.7 Manufacturability test 1

In order to study another solution to achieve higher accuracy in thicknesses, a skin panel (2016-150-P483-01) has been manufactured with a new infusion strategy: differential pressures. These are the deviations of the mean versus the nominal of this specimen:

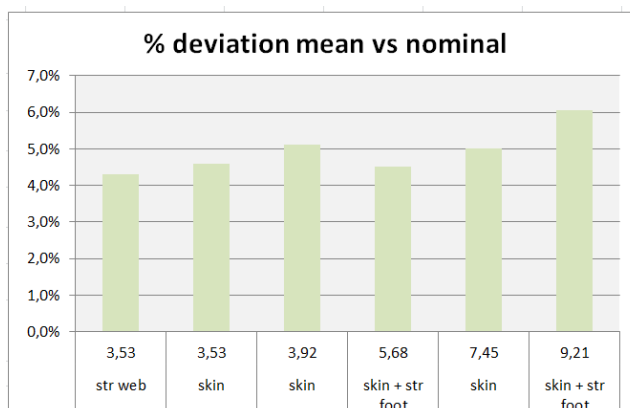


Figure 88 Deviation mean vs nominal of manufacturability test 1

In this manufacturability tests, the deviations of the mean with respect to the theoretical nominal are much lower than in the previous specimens and they are not so sensitive to absolute thickness variations.

	PREVIOUS TEST	MANUFAC. TEST
Mean	8,89%	4,93%
Highest value	15,65%	6,06%

Table 23 Mean and highest value of manufacturing test 1 in comparison to previous tests

The deviation of the mean versus the nominal value varies between 4,3 and 6,1 %, being the average deviation 4,93%, much lower than in the previous specimens considered for the study, where the mean was 8,89 % and it reaches values of 15,65%.

In summary, in this manufacturability test the deviations are remarkably lower. On the other hand, this new infusion strategy of differential pressures may not infuse properly. Zones with high porosity were found in this skin panel, which are not acceptable defects.

4.2.8 Manufacturability test 2

In order to study other possibilities of manufacturing and placing the tools and its possible influence, another skin panel (2016-150-P520-01) was manufactured. In this skin panel the caul plates used for stringers were placed outside the vacuum bag.

These are the results obtained for **stringer web thickness**:

CS A-3.1 (2016-150-P520-01)														
STRINGER WEB THICKNESS														
DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
		min		max					3 σ			2 σ		
nominal value	samples	min value	% desviation	max value	% desviation	mean	% deviation mean vs nominal	median	- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
4,31	50	4,18	3,04%	4,74	9,83%	4,48	3,90%	4,51	4,00	4,96	10,82%	4,16	4,80	7,14%

Table 24 Stringer web thickness data for manufacturability test 2

Comparing the data with the previous skin panels:

- Previously, the deviation of the maximum value with respect to the nominal value was 29,76% and in this new panel is 9,83%.

- The mean deviation with respect to the nominal is now 3,90% versus the previous 11,68%.

The results obtained from the new skin panel are significantly better, which was expected, as the manufacturing improvements affect directly to this property.

For **skin thickness**, these are the results obtained:

CS A-3.1 (2016-150-P520-01)														
SKIN THICKNESS														
DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
		min		max					3 σ			2 σ		
nominal value	samples	min value	% desviation	max value	% desviation	mean	% deviation mean vs nominal	median	- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
3,53	6	3,87	9,72%	4,21	19,33%	4,07	15,36%	4,07	3,71	4,44	8,99%	3,83	4,32	6,02%
3,92	18	4,24	8,24%	4,59	16,99%	4,41	12,50%	4,43	4,14	4,68	6,10%	4,23	4,59	4,06%
7,45	26	7,65	2,65%	8,13	9,20%	7,89	5,93%	7,89	7,54	8,24	4,41%	7,66	8,12	2,92%

Table 25 Skin thickness data for manufacturability test 2

Comparing the data with the previous skin panels:

- Mean of deviations with respect to the nominal is 11,26% versus the previous 11,94%.
- Deviations of new skin panel are only a bit lower, but it's reasonable considering that nominal thicknesses studied are higher.

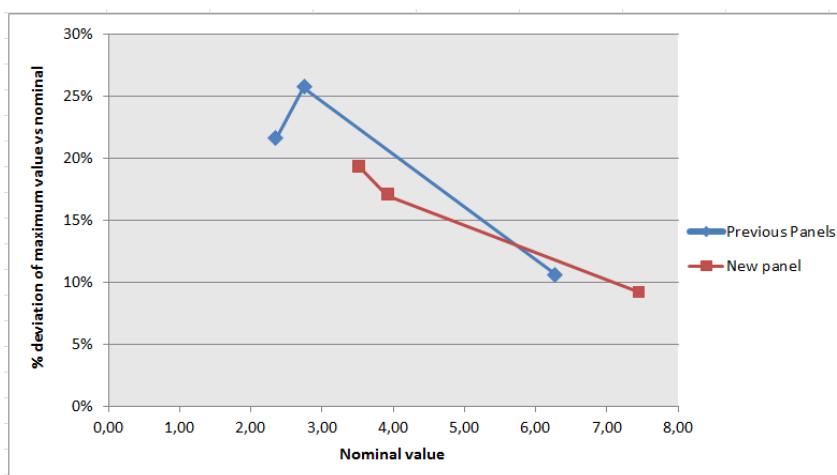


Figure 89 Skin thickness deviations of manufacturability test 2 vs previous panels

The results obtained from the new skin panel are very similar to the previous panels, which was expected, as the manufacturing improvements don't affect directly to skin thicknesses.

Finally, for **skin + stringer foot thickness** these are the results obtained:

CS A-3.1 (2016-150-P520-01)														
SKIN+STR FOOT THICKNESS														
DATA		MEASURED VALUES				AVERAGE VALUES			DEVIATIONS					
		min		max					3 σ			2 σ		
nominal value	samples	min value	% desviation	max value	% desviation	mean	% deviation mean vs nominal	median	- 3 σ	+ 3 σ	% deviation	- 2 σ	+ 2 σ	% deviation
5,68	10	6,21	9,24%	6,56	15,38%	6,32	11,19%	6,26	5,97	6,67	5,47%	6,09	6,55	3,64%
6,08	20	6,54	7,67%	6,90	13,53%	6,67	9,76%	6,64	6,38	6,96	4,32%	6,48	6,86	2,86%
9,60	20	10,08	4,98%	10,46	8,89%	10,27	6,93%	10,28	9,95	10,59	3,08%	10,06	10,48	2,04%

Table 26 Skin + stringer foot thickness data for manufacturability test 2

Comparing the data with the previous skin panels:

- The media of deviation of the maximum value with respect to the nominal value was 19,07% and in this new panel is 12,6%.
- The mean deviation with respect to the nominal is now 9,29% versus the previous 11,06%.

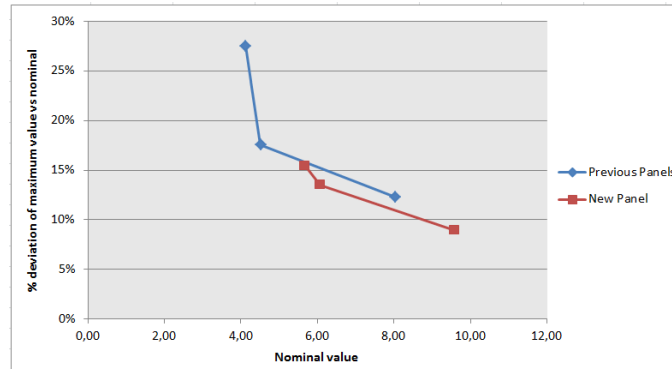


Figure 90 Skin + stringer foot thickness deviations of manufacturability test 2 vs previous panels

The results obtained from the new skin panel are slightly better, but not as good as expected, as the manufacturing improvements should possibly affect to this property, but instead the deviations have decreased considerably less than in stringer web thicknesses.

4.2.9 Conclusions thicknesses study

The results obtained in this statistical study show that the deviations are considerably higher than typical prepreg values. The percentages of thickness deviations from these test specimens are not suitable for a correct external wing box design, due to its effect in the stack-ups tolerances and the gaps between parts.

Although some additional manufacturability tests have been made, the results are not as good as necessary. Further trials are required.

For the moment a proposal of a coupon test matrix to evaluate MKDF of overthickness has been made. The matrix includes coupons with different values of overthickness which would be tested by different test methods.

4.3 Width of stringers

For width of stringers values there are available data from the four test specimens. There are approximately 200 data of width of stringers available. However some specific specimens have been taken out of the study because the laminates were cut narrower (the book effect was not considered).

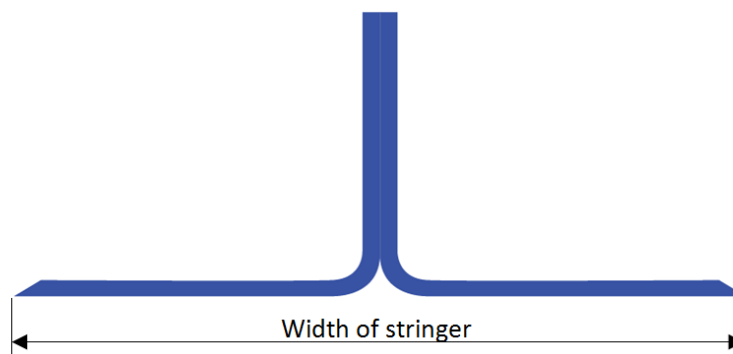


Figure 91 Width of stringer

In this case and the following, the data are not normally distributed, and the values of $\pm 2\sigma$ and $\pm 3\sigma$ are not being used. The objective now is to study the absolute deviations.

The acceptable deviation consider for this study is $\pm 1,5$ mm. These are the data obtained:

TEST INFORMATION			DATA		MEASURED VALUES				AVERAGE VALUES		
ID	description	property measured	nominal value (mm)	number of samples	min		max		mean (mm)	absolute deviation mean vs nom	median (mm)
					min value (mm)	absolute deviation	max value (mm)	absolute deviation			
CS A-2.1	Stringer Crippling	width of stringers	63,54	21	63,30	0,24	65,48	1,94	64,15	0,61	63,92
CS A-2.1	Stringer Crippling	width of stringers	67,77	18	68,10	0,33	70,89	3,12	69,05	1,28	69,03
CS A-2.2	Stringer Run-out	width of stringers	66,54	7	66,15	0,39	66,94	0,40	66,07	0,47	66,02
CS A-2.2	Stringer Run-out	width of stringers	79,24	21	77,78	1,46	80,68	1,44	79,40	0,16	79,62
CS A-3.1	Skin Panel	width of stringers	63,54	45	63,59	0,05	64,56	1,02	64,07	0,53	64,03
CS A-3.1	Skin Panel	width of stringers	80,74	30	80,86	0,12	82,97	2,23	81,81	1,07	81,90
CS A-3.3	Spar Web	width of stringers	44,31	36	43,76	0,55	45,60	1,29	44,73	0,42	44,68
CS A-3.3	Spar Web	width of stringers	50,54	18	49,61	0,93	51,57	1,03	50,50	0,04	50,37

Table 27 Width of stringer data

Nine of 196 measurements have deviations over 1,5 mm. Eight of these nine are from stringer crippling specimens. Only three have absolute deviations over 2 mm. High deviations are not common.

For these reasons, it has been considered reasonable to keep the tolerance of $\pm 1,5$ mm for width of stringers.

4.4 Stringer pitch

For stringer pitch, only data from skin panels reports are available, as they are the unique specimen type that includes more than one stringer. There are 80 data of stringer pitch available.

The pitch between stringers has been measured between the stringers faces. The local contributions of web thickness deviations with respect to nominal have not been considered.

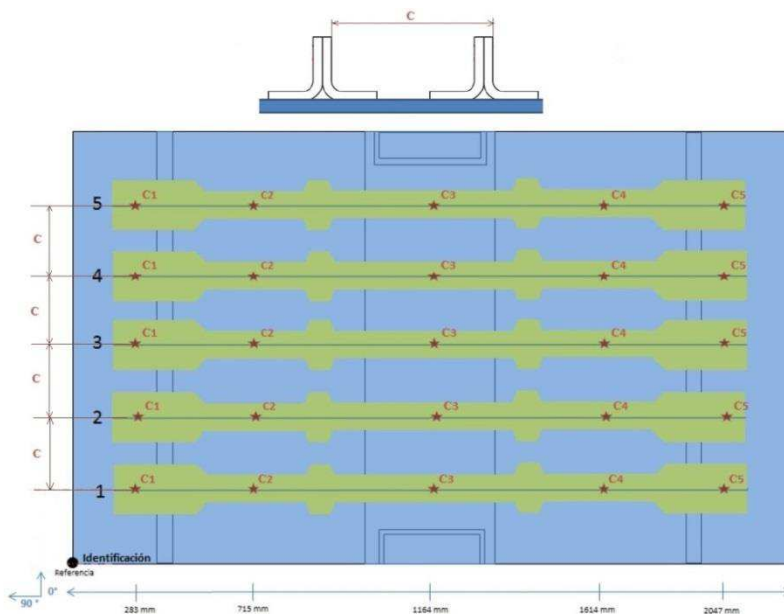


Figure 92 Stringer pitch

Only one nominal value has been studied, because the stringer pitch considered for these specimens is 110 mm. The acceptable deviation previously consider for this study is ± 3 mm for C2, C3 and C4 position (although the objective is to decrease it to $\pm 1,5$ mm) and ± 1 mm for C1 and C5,. These are the data obtained:

TEST INFORMATION			DATA			MEASURED VALUES				AVERAGE VALUES		
ID	description	property measured	nominal value (mm)	position	number of samples	min		max				
						min value (mm)	absolute deviation	max value (mm)	absolute deviation	mean (mm)	absolute deviation mean vs nom	median (mm)
CS A-3.1	Skin Panel	distance web str-end of skin	110	C1 and C5	32	108,95	1,05	110,90	0,90	109,93	0,07	109,89
CS A-3.1	Skin Panel	distance web str-end of skin	110	C2, C3 and C4	48	107,88	2,12	111,79	1,79	110,04	0,04	110,13

Table 28 Stringer pitch data

Only one value of C1 and C5 absolute deviations is over 1 mm (and it is 1.05). So the tolerance of ± 1 mm is validated.

Fifteen of 80 deviations are over 1 mm (and fourteen are in C2, C3 and C4 positions). Of these fifteen deviations only five are over 1,5 mm (with values of 1,56-1,79-1,90-1,93-2,12).

In this case there is only one deviation over 2 mm, and not too many over 1,5 mm, so it could be reasonable to decrease the tolerance to $\pm 1,5$ mm. However, this value is yet to be confirmed by Manufacturing Engineering.

4.5 Distance from stringer end to end of skin

In this study only skin panels reports are available, as they are the unique specimen type in which it make sense to do this study (other specimens are trimmed using the distance from stringer end as a reference). There are 30 data of distance from stringer end to end of skin available.

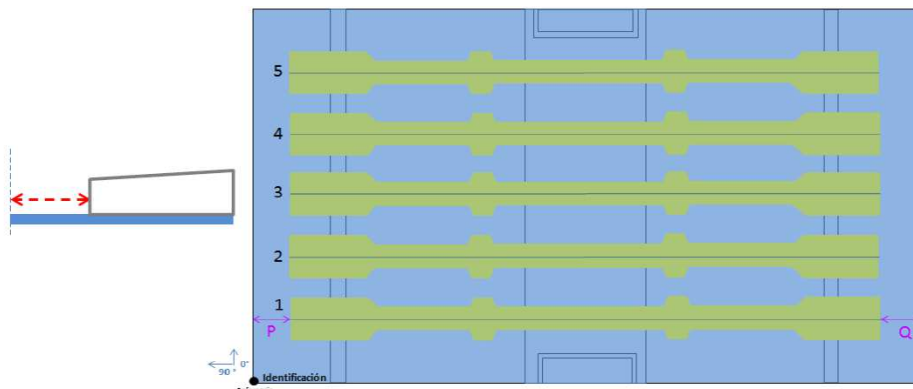


Figure 93 Distance from stringer end to end of skin

The acceptable deviation considered for this study is ± 1 mm for all measurements. These are the data obtained:

TEST INFORMATION				DATA		MEASURED VALUES				AVERAGE VALUES		
ID	description	position	property measured	nominal value (mm)	number of samples	min		max				
						min value (mm)	absolute deviation	max value (mm)	absolute deviation	mean (mm)	absolute deviation mean vs nom	median (mm)
CS A-3.1	Skin Panel	P	distance str end-end of skin	200	15	198,55	1,45	200,44	0,44	199,36	-0,64	199,18
CS A-3.1	Skin Panel	Q	distance str end-end of skin	200	15	198,71	1,29	202,23	2,23	199,82	-0,18	199,60

Table 29 Distance from stringer end to end of skin data

Eight of 30 measurements deviations are over 1 mm. These measurements are from the three specimens, positions P and Q and almost all stringers positions. Therefore the deviations out of tolerance are randomly distributed. The acceptance of this tolerance of ± 1 is yet to be considered, as more than 25 % of the measurements are out of tolerance.

5. CONCLUSIONS

The three objectives of this document previously proposed have been accomplished by doing:

- 1) A summary and evaluation of LRI technology and process steps, its differences with respect to prepreg technology, and its impact in Structural Design, including process and material special considerations.
- 2) A dimensional study of different properties from existing test specimens reports, with its corresponding conclusions and the acceptance (or refusal) of the previously considered tolerances.
- 3) A list of general and specific design principles for lifting surfaces covers manufactured with AFP + LRI process and materials.

The main conclusion of the study is that LRI is a technology for the future with a lot of improvement potential. Manufacturing improvements are required if there is an interest in developing this technology and using it widely, replacing traditional composites manufacturing processes.

Close attention must be paid to improving process control, for example, the resin infusion flow and different infusion techniques, that could accomplish better thicknesses accuracy, and in general a more reliable and repetitive LRI process.